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ORGANIC COMPOUND, LIGHT-EMITTING DEVICE INCLUDING THE SAME AND ELECTRONIC APPARATUS INCLUDING THE LIGHT-EMITTING DEVICE

Abstract

Embodiments provide a light-emitting device, an electronic apparatus including the light-emitting device, and an electronic equipment including the light-emitting device. The light-emitting device includes a first electrode, a second electrode facing the first electrode, and an interlayer between the first electrode and the second electrode and including an emission layer. The interlayer includes an organic compound including a boron (B) atom, wherein the organic compound satisfies Expression 1:

[00001] $k_{ISC} / k_{RISC} \leq 10$ [Expression1]

In Expression 1, k.sub.ISC is an intersystem crossing rate constant of the organic compound, and k.sub.RISC is a reverse intersystem crossing rate constant of the organic compound.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
[0001] This application claims priority to and benefits of Korean Patent Application No. 10-2024-0032221 under 35 U.S.C. § 119, filed on Mar. 6, 2024, in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.
BACKGROUND
1. Technical Field
[0002] Embodiments relate to an organic compound, a light-emitting device including the same, and an electronic apparatus including the light-emitting device.
2. Description of the Related Art
[0003] Light-emitting devices (for example, organic light-emitting devices) are self-emissive devices that have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of luminance, driving voltage, and response speed.
[0004] A light-emitting device may include a first electrode, a hole transport region, an emission layer, an electron transport region, and a second electrode that are arranged in that order. Holes injected from the first electrode may move toward the emission layer through the hole transport region. Electrons injected from the second electrode may move toward the emission layer through an electron injection layer in the electron transport region. Carriers, such as holes and electrons, may recombine in the emission layer to produce excitons. As the excitons transition from an excited state to a ground state, light may be generated.
[0005] It is to be understood that this background of the technology section is, in part, intended to provide useful background for understanding the technology. However, this background of the technology section may also include ideas, concepts, or recognitions that were not part of what was known or appreciated by those skilled in the pertinent art prior to a corresponding effective filing date of the subject matter disclosed herein.
SUMMARY
[0006] Embodiments include an organic compound with an accelerated reverse intersystem crossing rate and an accelerated radiative rate of a lowest singlet excited state, a light-emitting device including the organic compound with improved luminescence efficiency and lifespan, and an electronic apparatus with improved display quality including the light-emitting device.
[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the embodiments of the disclosure.
[0008] According to embodiments, a light-emitting device may include a first electrode, a second electrode facing the first electrode, and an interlayer between the first electrode and the second

electrode and including an emission layer, wherein the interlayer may include an organic compound that includes a boron (B) atom, and the organic compound may satisfy Expression 1:

[00002] $k_{ISC} / k_{RISC} \leq 10$ [Expression1]

[0009] In Expression 1, [0010] k.sub.ISC may be an intersystem crossing rate constant of the organic compound, and [0011] k.sub.RISC may be a reverse intersystem crossing rate constant of the organic compound.

[0012] In an embodiment, the interlayer may further include a hole transport host, an electron transport host, a sensitizer, or any combination thereof; and the organic compound, the hole transport host, the electron transport host, and the sensitizer may each be different from each other.

[0013] In an embodiment, the hole transport host and the electron transport host may form an exciplex.

[0014] In an embodiment, the hole transport host may be a compound including at least one carbazole group.

[0015] In an embodiment, the electron transport host may be a compound including at least one π electron-deficient nitrogen-containing 6-membered ring.

[0016] In an embodiment, the sensitizer may include a transition metal.

[0017] In an embodiment, the emission layer may include: the organic compound; and the hole transport host, the electron transport host, the sensitizer, or any combination thereof.

[0018] In an embodiment, the emission layer may emit blue light having a maximum emission wavelength in a range of about 450 nm to about 470 nm.

[0019] According to embodiments, an electronic apparatus may include the light-emitting device.

[0020] In an embodiment, the electronic apparatus may further include a thin-film transistor electrically connected to the light-emitting device.

[0021] In an embodiment, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof.

[0022] According to embodiments, an electronic equipment may the light-emitting device, wherein the electronic equipment may be a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.

[0023] According to embodiments, an organic compound may include a boron (B) atom, wherein the organic compound may satisfy Expression 1, which is explained herein.

[0024] In an embodiment, the organic compound may further satisfy at least one of Expressions 2 and 3, which are explained herein.

[0025] In an embodiment, the organic compound may further satisfy at least one of Expressions 4 and 5, which are explained herein.

[0026] In an embodiment, the organic compound may further satisfy Expression 6, which is explained herein.

[0027] In an embodiment, the organic compound may further satisfy Expression 7, which is explained herein.

[0028] In an embodiment, the organic compound may further satisfy Expression 8, which is explained herein.

[0029] In an embodiment, the organic compound may further include at least one nitrogen atom.

[0030] In an embodiment, the organic compound may include a heterocyclic group containing a boron atom and a nitrogen atom each as a ring-forming atom.

[0031] It is to be understood that the embodiments above are described in a generic and explanatory sense only and not for the purposes of limitation, and the disclosure is not limited to the embodiments described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] The accompanying drawings are included to provide a further understanding of the embodiments, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and principles thereof. The above and other aspects and features of the disclosure will become more apparent by describing in detail embodiments thereof with reference to the accompanying drawings, in which:

[0033] FIG. **1** is a graph of energy levels at room temperature of an organic compound according to an embodiment;

[0034] FIG. **2** is a schematic cross-sectional view of a light-emitting device according to an embodiment;

[0035] FIG. **3** is a schematic cross-sectional view of an electronic apparatus according to an embodiment;

[0036] FIG. **4** is a schematic cross-sectional view of an electronic apparatus according to another embodiment;

[0037] FIG. **5** is a schematic perspective view of an electronic equipment including a light-emitting device according to an embodiment;

[0038] FIG. **6** is a schematic perspective view of an exterior of a vehicle as an electronic equipment including a light-emitting device according to an embodiment; and

[0039] FIGS. **7**A to **7**C are each a schematic diagram of an interior of a vehicle according to embodiments.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0040] The disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments are shown. This disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art.

[0041] In the drawings, the sizes, thicknesses, ratios, and dimensions of the elements may be exaggerated for ease of description and for clarity. Like reference numbers and reference characters refer to like elements throughout.

[0042] In the specification, it will be understood that when an element (or region, layer, part, etc.) is referred to as being "on", "connected to", or "coupled to" another element, it can be directly on, connected to, or coupled to the other element, or one or more intervening elements may be present therebetween. In a similar sense, when an element (or region, layer, part, etc.) is described as "covering" another element, it can directly cover the other element, or one or more intervening elements may be present therebetween.

[0043] In the specification, when an element is "directly on", "directly connected to", or "directly coupled to" another element, there are no intervening elements present. For example, "directly on" may mean that two layers or two elements are disposed without an additional element such as an adhesion element therebetween.

[0044] In the specification, the expressions used in the singular such as "a", "an", and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0045] In the specification, the term "and/or" includes any and all combinations of one or more of the associated listed items. For example, "A and/or B" may be understood to mean "A, B, or A and B". The terms "and" and "or" may be used in the conjunctive or disjunctive sense and may be

understood to be equivalent to “and/or”.
[0046] In the specification and the claims, the term “at least one of” is intended to include the meaning of “at least one selected from the group consisting of” for the purpose of its meaning and interpretation. For example, “at least one of A, B, and C” may be understood to mean A only, B only, C only, or any combination of two or more of A, B, and C, such as ABC, ACC, BC, or CC. When preceding a list of elements, the term, “at least one of”, modifies the entire list of elements and does not modify the individual elements of the list.
[0047] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, a first element could be termed a second element without departing from the teachings of the disclosure. Similarly, a second element could be termed a first element, without departing from the scope of the disclosure.
[0048] The spatially relative terms “below”, “beneath”, “lower”, “above”, “upper”, or the like, may be used herein for ease of description to describe the relations between one element or component and another element or component as illustrated in the drawings. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the drawings. For example, in the case where a device illustrated in the drawing is turned over, the device positioned “below” or “beneath” another device may be placed “above” another device. Accordingly, the illustrative term “below” may include both the lower and upper positions. The device may also be oriented in other directions and thus the spatially relative terms may be interpreted differently depending on the orientations.
[0049] The terms “about” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the recited value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the recited quantity (i.e., the limitations of the measurement system). For example, “about” may mean within one or more standard deviations, or within ±20%, ±10%, or ±5% of the stated value.
[0050] It should be understood that the terms “comprises”, “comprising”, “includes”, “including”, “have”, “having”, “contains”, “containing”, and the like are intended to specify the presence of stated features, integers, steps, operations, elements, components, or combinations thereof in the disclosure, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof.
[0051] Unless otherwise defined or implied herein, all terms (including technical and scientific terms) used have the same meaning as commonly understood by those skilled in the art to which this disclosure pertains. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an ideal or excessively formal sense unless clearly defined in the specification.
[0052] According to an embodiment, an organic compound may include a boron (B) atom, wherein the organic compound may satisfy Expression 1:
[00003] $k_{\mathrm{ISC}} / k_{\mathrm{RISC}} \leq 10$ [Expression1]
[0053] In Expression 1, [0054] k.sub.ISC may be an intersystem crossing rate constant of the organic compound, and [0055] k.sub.RISC may be a reverse intersystem crossing rate constant of the organic compound.
[0056] Therefore, according to an embodiment, the organic compound is clearly different from a compound that does not include a boron (B) atom.
[0057] According to an embodiment, k.sub.ISC/k.sub.RISC may be less than 10. For example, k.sub.ISC/k.sub.RISC may be equal to or less than 9.5. For example, k.sub.ISC/k.sub.RISC may be less than 9.5. For example, k.sub.ISC/k.sub.RISC may be equal to or less than 9.0. For example, k.sub.ISC/k.sub.RISC may be less than 9.0. For example, k.sub.ISC/k.sub.RISC may be equal to or

less than 8.5. For example, k.sub.ISC/k.sub.RISC may be less than 8.5. For example, k.sub.ISC/k.sub.RISC may be equal to or less than 8.0. For example, k.sub.ISC/k.sub.RISC may be less than 8.0.

[0058] According to an embodiment, k.sub.ISC/k.sub.RISC may be equal to or greater than 1.0. For example, k.sub.ISC/k.sub.RISC may be greater than 1.0. For example, k.sub.ISC/k.sub.RISC may be equal to or greater than 2.0. For example, k.sub.ISC/k.sub.RISC may be greater than 2.0. For example, k.sub.ISC/k.sub.RISC may be equal to or greater than 3.0. For example, k.sub.ISC/k.sub.RISC may be greater than 3.0. For example, k.sub.ISC/k.sub.RISC may be equal to or greater than 4.0. For example, k.sub.ISC/k.sub.RISC may be greater than 4.0. For example, k.sub.ISC/k.sub.RISC may be equal to or greater than 5.0. For example, k.sub.ISC/k.sub.RISC may be greater than 5.0.

[0059] According to an embodiment, k.sub.ISC/k.sub.RISC may be in a range of about 1.0 to about 10. For example, k.sub.ISC/k.sub.RISC may be in a range of about 1.5 to about 9.5. For example, k.sub.ISC/k.sub.RISC may be in a range of about 2.0 to about 9.0. For example, k.sub.ISC/k.sub.RISC may be in a range of about 2.5 to about 8.5. For example, k.sub.ISC/k.sub.RISC may be in a range of about 3.0 to about 8.0. For example, k.sub.ISC/k.sub.RISC may be in a range of about 4.0 to about 8.0.

[0060] FIG. **1** is a graph of energy levels at room temperature of an organic compound according to an embodiment. Referring to FIG. **1**, the organic compound may have a ground state (S.sub.0), a lowest singlet excited state (S.sub.1), and a lowest triplet excited state (T.sub.1). A difference between an energy level of the lowest singlet excited state (S.sub.1) and an energy level of the lowest triplet excited state (T.sub.1) may be expressed as ΔE.sub.ST.

[0061] Excitons in the lowest singlet excited state (S.sub.1) may undergo radiation, non-radiative decay, and intersystem crossing (ISC) to transition to the lowest triplet excited state (T.sub.1). Excitons in the lowest triplet excited state (T.sub.1) may undergo non-radiative decay and reverse intersystem crossing (RISC) to transition to the lowest singlet excited state (S.sub.1).

[0062] A radiative rate constant of the lowest singlet excited state (S.sub.1) may be expressed as k.sub.r.sup.S. A non-radiative decay rate constant of the lowest singlet excited state (S.sub.1) may be expressed as k.sub.nr.sup.S. An intersystem crossing rate constant of the lowest singlet excited state (S.sub.1) may be expressed as k.sub.ISC. A non-radiative decay rate constant of the lowest triplet excited state (T.sub.1) may be expressed as k.sub.nr.sup.T. A reverse intersystem crossing rate constant of the lowest triplet excited state (T.sub.1) may be expressed as k.sub.RISC.

[0063] Prompt fluorescence (PF) may refer to where an exciton transferred to the lowest singlet excited state (S.sub.1) of the organic compound does not undergo an intersystem crossing to the lowest triplet excited state (T.sub.1), moves to the ground state (S.sub.0), and emits light.

[0064] Delayed fluorescence (DF) may refer to where an exciton transferred to the lowest singlet excited state (S.sub.1) of the organic compound undergoes an intersystem crossing to the lowest triplet excited state (T.sub.1) and undergoes a reverse intersystem crossing to the lowest singlet excited state (S.sub.1), moves to the ground state (S.sub.0), and emits light.

[0065] According to an embodiment, the organic compound may include at least one nitrogen (N) atom. For example, the organic compound may include a heterocyclic group containing a boron atom and a nitrogen atom each as a ring-forming atom.

[0066] According to an embodiment, the organic compound may include at least one oxygen (O) atom. For example, the organic compound may include a heterocyclic group containing a boron atom and an oxygen atom each as a ring-forming atom.

[0067] According to an embodiment, the organic compound may further satisfy at least one of Expressions 2 and 3:

[00004] $k_{ISC} \geq 10^5 s^{-1}$ [Expression2] $k_{RISC} \geq 10^5 s^{-1}$. [Expression3]

[0068] In an embodiment, the organic compound may satisfy both Expression 2 and Expression 3.

[0069] According to an embodiment, k.sub.ISC may be greater than 10.sup.5 s.sup.−1. For example, k.sub.ISC may be equal to or greater than 5×10.sup.5 s.sup.−1. For example, k.sub.ISC may be greater than 5×10.sup.5 s.sup.−1. For example, k.sub.ISC may be equal to or greater than 10.sup.6 s.sup.−1. For example, k.sub.ISC may be greater than 10.sup.6 s.sup.−1. For example, k.sub.ISC may be equal to or greater than 2.5×10.sup.6 s.sup.−1. For example, k.sub.ISC may be greater than 2.5×10.sup.6 s.sup.−1. In an embodiment, k.sub.ISC may be equal to or less than 5×10.sup.8 s.sup.−1. For example, k.sub.ISC may be less than 5×10.sup.8 s.sup.−1. For example, k.sub.ISC may be equal to or less than 10.sup.8 s.sup.−1. For example, k.sub.ISC may be less than 10.sup.8 s.sup.−1. For example, k.sub.ISC may be equal to or less than 5×10.sup.7 s.sup.−1. For example, k.sub.ISC may be less than 5×10.sup.7 s.sup.−1. For example, k.sub.ISC may be equal to or less than 10.sup.7 s.sup.−1. For example, k.sub.ISC may be less than 10.sup.7 s.sup.−1. For example, k.sub.ISC may be equal to or less than 5×10.sup.6 s.sup.−1. For example, k.sub.ISC may be less than 5×10.sup.6 s.sup.−1. For example, k.sub.ISC may be equal to or less than 3×10.sup.6 s.sup.−1.

[0070] According to an embodiment, k.sub.RISC may be greater than 10.sup.4 s.sup.−1. For example, k.sub.RISC may be equal to or greater than 5×10.sup.4 s.sup.−1. For example, k.sub.RISC may be greater than 5×10.sup.4 s.sup.−1. For example, k.sub.RISC may be equal to or greater than 10.sup.5 s.sup.−1. For example, k.sub.RISC may be greater than 10.sup.5 s.sup.−1. For example, k.sub.RISC may be equal to or greater than 2×10.sup.5 s.sup.−1. For example, k.sub.RISC may be greater than 2×10.sup.5 s.sup.−1. In an embodiment, k.sub.RISC may be equal to or less than 5×10.sup.7 s.sup.−1. For example, k.sub.RISC may be less than 5×10.sup.7 s.sup.−1. For example, k.sub.RISC may be equal to or less than 10.sup.7 s.sup.−1. For example, k.sub.RISC may be less than 10.sup.7 s.sup.−1. For example, k.sub.RISC may be equal to or less than 5×10.sup.6 s.sup.−1. For example, k.sub.RISC may be less than 5×10.sup.6 s.sup.−1. For example, k.sub.RISC may be equal to or less than 10.sup.6 s.sup.−1. For example, k.sub.RISC may be less than 10.sup.6 s.sup.−1. For example, k.sub.RISC may be equal to or less than 5×10.sup.5 s.sup.−1. For example, k.sub.RISC may be less than 5×10.sup.5 s.sup.−1.

[0071] According to an embodiment, the organic compound may further satisfy at least one of the Expressions 4 and 5:

[00005] $k_r^S > k_{\mathrm{ISC}}$ [Expression4] $k_r^S > k_{\mathrm{RISC}}$ [Expression5]

[0072] In Expressions 4 and 5, [0073] k.sub.r.sup.S may be a radiative rate constant of the lowest singlet excited state of the organic compound.

[0074] In an embodiment, the organic compound may satisfy both Expression 4 and Expression 5.

[0075] According to an embodiment, the organic compound may further satisfy Expression 6:

[00006] $k_r^S \geq 10^7\, s^{-1}$ [Expression6]

[0076] In Expression 6, [0077] k.sub.r.sup.S may be a radiative rate constant of the lowest singlet excited state of the organic compound.

[0078] According to an embodiment, k.sub.r.sup.S may be greater than 10.sup.6 s.sup.−1. For example, k.sub.r.sup.S may be equal to or greater than 5×10.sup.6 s.sup.−1. For example, k.sub.r.sup.S may be greater than 5×10.sup.6 s.sup.−1. For example, k.sub.r.sup.S may be equal to or greater than 10.sup.7 s.sup.−1. For example, k.sub.r.sup.S may be greater than 10.sup.7 s.sup.−1. For example, k.sub.r.sup.S may be equal to or greater than 5×10.sup.7 s.sup.−1. For example, k.sub.r.sup.S may be greater than 5×10.sup.7 s.sup.−1. For example, k.sub.r.sup.S may be equal to or greater than 8×10.sup.7 s.sup.−1.

[0079] According to an embodiment, k.sub.r.sup.S may be equal to or less than 5×10.sup.9 s.sup.−1. For example, k.sub.r.sup.S may be less than 5×10.sup.9 s.sup.−1. For example, k.sub.r.sup.S may be equal to or less than 10.sup.9 s.sup.−1. For example, k.sub.r.sup.S may be less than 10.sup.9 s.sup.−1. For example, k.sub.r.sup.S may be equal to or less than 5×10.sup.8 s.sup.−1. For example, k.sub.r.sup.S may be less than 5×10.sup.8 s.sup.−1. For example, k.sub.r.sup.S may be equal to or less than 10.sup.8 s.sup.−1. For example, k.sub.r.sup.S may be less than 10.sup.8 s.sup.−1.

[0080] In an embodiment, the organic compound may satisfy Expression 2-1:

[00007] $k_r^S \geq 10^7 s^{-1} > k_{ISC} \geq 10^5 s^{-1}$. [Expression2 - 1]

[0081] In an embodiment, the organic compound may satisfy Expression 3-1:

[00008] $k_r^S \geq 10^7 s^{-1} > k_{RISC} \geq 10^5 s^{-1}$. [Expression3 - 1]

[0082] According to an embodiment, the organic compound may further satisfy Expression 7:

[00009] $_{PF} / (_{PF} + _{DF}) \geq 90\%$ [Expression7]

[0083] In Expression 7, [0084] ϕ.sub.PF may be a prompt fluorescence photoluminescence quantum yield (PF PLQY) of the organic compound, and [0085] ϕ.sub.DF may be a delayed fluorescence photoluminescence quantum yield (DF PLQY) of the organic compound.

[0086] The photoluminescence quantum yield (PLQY) may be measured for a film including the organic compound. For example, the film may include the organic compound and a host (for example, a second compound and/or a third compound to be described below). The PLQY of the organic compound may be equal to the sum of ϕ.sub.PF and ϕ.sub.DF.

[0087] According to an embodiment, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 90.1%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 90.2%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 90.3%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 90.4%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 90.5%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 90.6%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 91%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or greater than 91.5%.

[0088] According to an embodiment, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be 99%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 98%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 97%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 96%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 95%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 94%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 93%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 92%. For example, ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) may be equal to or less than 91%.

[0089] According to an embodiment, the organic compound may further satisfy Expression 8:

[00010] $E_{ST} \leq 0.15\text{eV}$ [Expression8]

[0090] In Expression 8, [0091] ΔE.sub.ST may be a difference between a lowest singlet excited state (S.sub.1) energy level of the organic compound and a lowest triplet excited state (T.sub.1) energy level of the organic compound.

[0092] According to an embodiment, ΔE.sub.ST may be equal to or less than 0.14 eV. For example, ΔE.sub.ST may be equal to or less than 0.13 eV. For example, ΔE.sub.ST may be equal to or less than 0.12 eV. For example, ΔE.sub.ST may be equal to or less than 0.11 eV. For example, ΔE.sub.ST may be equal to or less than 0.10 eV.

[0093] According to an embodiment, the organic compound may be any one of Compounds DFD1 to DFD17:

##STR00001## ##STR00002##

[0094] According to an embodiment, the organic compound may include 6 to 10 heteroatoms. For example, the organic compound may include 6 to 8 heteroatoms, 6 to 7 heteroatoms, 7 to 8 heteroatoms, or 7 heteroatoms. In the specification, a heteroatom may be any atom other than a carbon atom and a hydrogen atom. Examples of a heteroatom may include B, N, and O.

[0095] According to an embodiment, the organic compound may include 2 to 4 boron atoms. For example, the organic compound may include 2 to 3 boron atoms, or 2 boron atoms.
[0096] According to an embodiment, the organic compound may include 2 to 4 oxygen atoms. For example, the organic compound may include 2 to 3 oxygen atoms, or 2 oxygen atoms.
[0097] According to an embodiment, the organic compound may include 2 to 4 nitrogen atoms. For example, the organic compound may include 2 to 3 nitrogen atoms, 3 to 4 nitrogen atoms, or 3 nitrogen atoms.
[0098] According to an embodiment, a light-emitting device may include: a first electrode; a second electrode facing the first electrode; and an interlayer between the first electrode and the second electrode and including an emission layer, wherein the interlayer may include an organic compound that includes a boron (B) atom, and the organic compound may satisfy Expression 1, as defined herein.
[0099] According to an embodiment, the interlayer may further include a hole transport host, an electron transport host, a sensitizer, or any combination thereof; and the organic compound, the hole transport host, the electron transport host, and the sensitizer may be different from each other. In an embodiment, the dopant may include the organic compound. For example, the organic compound may be a fluorescent dopant or delayed fluorescence dopant.
[0100] According to an embodiment, the hole transport host and the electron transport host may form an exciplex.
[0101] According to an embodiment, the hole transport host may be a compound including at least one carbazole group.
[0102] In an embodiment, the hole transport host may be any one of Compounds HTH1 to HTH17:
##STR00003## ##STR00004##
[0103] According to an embodiment, the electron transport host may be a compound including at least one π electron-deficient nitrogen-containing 6-membered ring. For example, the electron transport host may include a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or any combination thereof.
[0104] In an embodiment, the electron transport host may be any one of Compounds ETH1 to ETH16:
##STR00005## ##STR00006##
[0105] According to an embodiment, the sensitizer may include a transition metal. For example, the sensitizer may include platinum (Pt).
[0106] In an embodiment, the sensitizer may be any one of Compounds PD1 to PD39:
##STR00007## ##STR00008## ##STR00009## ##STR00010## ##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016## ##STR00017##
[0107] According to an embodiment, the emission layer may include: the organic compound; and the hole transport host, the electron transport host, the sensitizer, or any combination thereof.
[0108] According to an embodiment, the emission layer may emit blue light having a maximum emission wavelength in a range of about 450 nm to about 470 nm.
[0109] According to an embodiment, the light-emitting device may include a layer that includes: the organic compound; and the hole transport host, the electron transport host, the sensitizer, or any combination thereof. For example, the aforementioned “layer” may include a mixture that includes: the organic compound; and the hole transport host, the electron transport host, the sensitizer, or any combination thereof. For example, the “layer” is clearly distinguished from a double-layered structure consisting of: a first layer including the organic compound; and a second layer including the hole transport host, the electron transport host, the sensitizer, or any combination thereof. For example, the “layer” may be the emission layer.
[0110] According to an embodiment, an electronic apparatus may include the light-emitting device.
[0111] According to an embodiment, the electronic apparatus may further include a thin-film transistor electrically connected to the light-emitting device.

[0112] In an embodiment, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof.
[0113] According to an embodiment, an electronic equipment may include the light-emitting device, wherein the electronic equipment may be a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.
[0114] Compounds of the related art that contain a boron atom and a nitrogen atom may have multiple resonance (MR) characteristics by having a highest occupied molecular orbital (HOMO) region and a lowest unoccupied molecular orbital (LUMO) region. These compounds may have short charge transfer (short CT) characteristics because the distance between the HOMO region and the LUMO region is relatively close. Accordingly, these compounds may have a large oscillator strength f value, but a relatively small reverse intersystem crossing rate constant (for example, k.sub.RISC≤10.sup.4 s.sup.−1).
[0115] However, among such compounds that contain a boron atom and a nitrogen atom, organic compounds that appropriately include both a portion where the distance between the HOMO region and LUMO region are relatively close and a portion where the distance between the HOMO region and LUMO region are relatively long may appropriately have both short charge transfer (short CT) characteristics and long charge transfer (long CT) characteristics, and as a result, such organic compounds may satisfy at least Expression 1 among Expressions 1 to 8. By containing a boron atom and by satisfying Expression 1, such organic compounds may have a relatively high k.sub.r.sup.S and a relatively high k.sub.RISC, as compared to boron-containing compounds that do not satisfy Expression 1. Such organic compounds may have a similar oscillator strength f value as that of the boron-containing compounds that do not satisfy Expression 1.
[0116] Therefore, when a boron-containing organic compound satisfies Expression 1, triplet excitons may be harvested into singlet excitons relatively quickly, and the singlet excitons may transition relatively quickly to the ground state and emit light. Therefore, a light-emitting device employing the organic compound may have high luminescence efficiency and a long lifespan, and as a result, an electronic apparatus employing the light-emitting device may have improved display quality.
[0117] For example, the organic compound may have short charge transfer (short CT) characteristics in which the distance between the HOMO region and the LUMO region in each 6-membered ring group is relatively close and also may have long charge transfer (long CT) characteristics because the LUMO region of the moiety composed of the left boron atom and two oxygen atoms adjacent thereto and the HOMO region of the moiety composed of the right boron atom and three nitrogen atoms adjacent thereto are relatively far apart, such as in Compounds DFD13 to DFD17:
##STR00018## ##STR00019##
Description of FIG. **2**
[0118] FIG. **2** is a schematic cross-sectional view of a light-emitting device **10** according to an embodiment. The light-emitting device **10** may include a first electrode **110**, an interlayer, and a second electrode **150**. The interlayer may include a hole transport region **120**, an emission layer **130**, and an electron transport region **140**.
[0119] Hereinafter, the structure of the light-emitting device **10** according to an embodiment and a method of manufacturing the light-emitting device **10** will be described with reference to FIG. **2**.

[First Electrode **110**]

[0120] In FIG. **2**, a substrate may be further included under the first electrode **110** or on the second electrode **150**. The substrate may be a glass substrate or a plastic substrate. In an embodiment, the substrate may be a flexible substrate. For example, a flexible substrate may include plastics with excellent heat resistance and durability, such as polyimide, polyethylene terephthalate (PET), polycarbonate, polyethylene naphthalate, polyarylate (PAR), polyetherimide, or any combination thereof.

[0121] The first electrode **110** may be formed by depositing or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, a high-work function material that facilitates injection of holes may be used as a material for forming the first electrode **110**.

[0122] The first electrode **110** may be a reflective electrode, a transflective electrode, or a transmissive electrode. When the first electrode **110** is a transmissive electrode, a material for forming the first electrode **110** may include indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO_2), zinc oxide (ZnO), or any combination thereof. When the first electrode **110** is a transflective electrode or a reflective electrode, a material for forming the first electrode **110** may include magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), or any combination thereof.

[0123] The first electrode **110** may have a structure consisting of a single layer or a structure including multiple layers. For example, the first electrode **110** may have a three-layered structure of ITO/Ag/ITO.

[Interlayer]

[0124] The interlayer may be arranged on the first electrode **110**. The interlayer may include a hole transport region **120**, an emission layer **130**, and an electron transport region **140**.

[0125] The interlayer may include various organic materials, a metal-containing compound such as an organometallic compound, an inorganic material such as quantum dots, or the like.

[0126] In an embodiment, the interlayer may include two or more emitting units stacked between the first electrode **110** and the second electrode **150**, and at least one charge generation layer between adjacent units among the two or more emitting units. When the interlayer includes the two or more emitting units and the at least one charge generation layer as described above, the light-emitting device **10** may be a tandem light-emitting device.

[Hole Transport Region **120**]

[0127] The hole transport region **120** may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer including different materials, or a structure including multiple layers including different materials.

[0128] The hole transport region **120** may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof.

[0129] In embodiments, the hole transport region **120** may have a multi-layered structure including a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein the layers of each structure may be stacked from the first electrode **110** in its respective stated order, but the structure of the hole transport region **120** is not limited thereto.

[0130] In embodiments, the hole transport region **120** may include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof:

##STR00020##

[0131] In Formulae 201 and 202, [0132] L_{201} to L_{204} may each independently be a C_3-C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C_1-C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a},

[0133] L.sub.205 may be *—O—*′, *—S—*′, *—N(Q.sub.201)-*′, a C.sub.1-C.sub.20 alkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0134] xa1 to xa4 may each independently be an integer from 0 to 5, [0135] xa5 may be an integer from 1 to 10, [0136] R.sub.201 to R.sub.204 and Q.sub.201 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0137] R.sub.201 and R.sub.202 may optionally be linked to each other via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 polycyclic group (for example, a carbazole group, etc.) unsubstituted or substituted with at least one R.sub.10a (for example, see Compound HT16, etc.), [0138] R.sub.203 and R.sub.204 may optionally be linked to each other via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 polycyclic group unsubstituted or substituted with at least one R.sub.10a, and [0139] na1 may be an integer from 1 to 4.

[0140] In embodiments, the compound represented by Formula 201 and the compound represented by Formula 202 may each independently include at least one of groups represented by Formulae CY201 to CY217:

##STR00021## ##STR00022## ##STR00023## ##STR00024## ##STR00025## ##STR00026## ##STR00027##

[0141] In Formulae CY201 to CY217, R.sub.10b and R.sub.10c may each independently be the same as described with respect to R.sub.10a, ring CY201 to ring CY204 may each independently be a C.sub.3-C.sub.20 carbocyclic group or a C.sub.1-C.sub.20 heterocyclic group, and at least one hydrogen in Formulae CY201 to CY217 may be unsubstituted or substituted with R.sub.10a as described herein.

[0142] According to an embodiment, in Formulae CY201 to CY217, ring CY201 to ring CY.sub.204 may each independently be a benzene group, a naphthalene group, a phenanthrene group, or an anthracene group.

[0143] According to an embodiment, the compound represented by Formula 201 and the compound represented by Formula 202 may each independently include at least one of groups represented by Formulae CY201 to CY203.

[0144] According to an embodiment, the compound represented by Formula 201 may include at least one of the groups represented by Formulae CY201 to CY203 and at least one of groups represented by Formulae CY204 to CY217.

[0145] According to an embodiment, in Formula 201, xa1 may be 1, R.sub.201 may be a group represented by one of Formulae CY201 to CY203, xa2 may be 0, and R.sub.202 may be a group represented by one of Formulae CY204 to CY207.

[0146] According to an embodiment, the compound represented by Formula 201 and the compound represented by Formula 202 may each not include the groups represented by Formulae CY201 to CY203.

[0147] According to an embodiment, the compound represented by Formula 201 and the compound represented by Formula 202 may each not include the groups represented by Formulae CY201 to CY203, and may each independently include at least one of the groups represented by Formulae CY204 to CY217.

[0148] In an embodiment, the compound represented by Formula 201 and the compound represented by Formula 202 may each not include the groups represented by Formulae CY201 to CY217.

[0149] In an embodiment, the hole transport region **120** may include one of Compounds HT1 to HT46, m-MTDATA, TDATA, 2-TNATA, NPB (NPD), β-NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTPD, 4,4′,4″-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), or any combination thereof: ##STR00028## ##STR00029## ##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035## ##STR00036## ##STR00037##
[0150] A thickness of the hole transport region **120** may be in a range of about 50 Å to about 10,000 Å. For example, the thickness of the hole transport region **120** may be in a range of about 100 Å to about 4,000 Å. When the hole transport region **120** includes a hole injection layer, a hole transport layer, or any combination thereof, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å. For example, the thickness of the hole injection layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the hole transport layer may be in a range of about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region **120**, the hole injection layer, and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.
[0151] The emission auxiliary layer may increase light emission efficiency by compensating for an optical resonance distance according to a wavelength of light emitted from the emission layer **130**. The electron blocking layer may prevent electron leakage from the emission layer **130** to the hole transport region **120**. Materials that may be included in the hole transport region **120** may be included in the emission auxiliary layer and the electron blocking layer.
[p-Dopant]
[0152] The hole transport region **120** may include, in addition to the materials as described above, a charge-generation material for the improvement of conductive properties. The charge-generation material may be uniformly or non-uniformly dispersed in the hole transport region **120** (for example, in the form of a single layer consisting of a charge-generation material).
[0153] The charge-generation material may be, for example, a p-dopant.
[0154] For example, the p-dopant may have a lowest unoccupied molecular orbital (LUMO) energy level equal to or less than about −3.5 eV.
[0155] According to an embodiment, the p-dopant may include a quinone derivative, a cyano group-containing compound, a compound including element EL**1** and element EL**2**, or any combination thereof.
[0156] Examples of a quinone derivative may include TCNQ, F4-TCNQ, and the like.
[0157] Examples of a cyano group-containing compound may include HAT-CN and a compound represented by Formula 221:
##STR00038##
[0158] In Formula 221, [0159] R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, and [0160] at least one of R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each substituted with: a cyano group; —F; —Cl; —Br; —I; a C.sub.1-C.sub.20 alkyl group substituted with a cyano group, —F, —Cl, —Br, —I, or any combination thereof; or any combination thereof.
[0161] In the compound including element EL**1** and element EL**2**, element EL**1** may be a metal, a metalloid, or any combination thereof, and element EL**2** may be a non-metal, a metalloid, or any combination thereof.
[0162] Examples of a metal may include: an alkali metal (for example, lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), etc.); an alkaline earth metal (for example, beryllium

(Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), etc.); a transition metal (for example, titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), manganese (Mn), technetium (Tc), rhenium (Re), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), etc.); a post-transition metal (for example, zinc (Zn), indium (In), tin (Sn), etc.); a lanthanide metal (for example, lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), etc.); and the like.
[0163] Examples of a metalloid may include silicon (Si), antimony (Sb), tellurium (Te), and the like.
[0164] Examples of a non-metal may include oxygen (O), a halogen (for example, F, Cl, Br, I, etc.), and the like.
[0165] Examples of a compound including element EL**1** and element EL**2** may include a metal oxide, a metal halide (for example, a metal fluoride, a metal chloride, a metal bromide, a metal iodide, etc.), a metalloid halide (for example, a metalloid fluoride, a metalloid chloride, a metalloid bromide, a metalloid iodide, etc.), a metal telluride, or any combination thereof.
[0166] Examples of a metal oxide may include a tungsten oxide (for example, WO, W.sub.2O.sub.3, WO.sub.2, WO.sub.3, W.sub.2O.sub.5, etc.), a vanadium oxide (for example, VO, V.sub.2O.sub.3, VO.sub.2, V.sub.2O.sub.5, etc.), a molybdenum oxide (for example, MoO, Mo.sub.2O.sub.3, MoO.sub.2, MoO.sub.3, Mo.sub.2O.sub.5, etc.), a rhenium oxide (for example, ReO.sub.3, etc.), and the like.
[0167] Examples of a metal halide may include an alkali metal halide, an alkaline earth metal halide, a transition metal halide, a post-transition metal halide, a lanthanide metal halide, and the like.
[0168] Examples of an alkali metal halide may include LiF, NaF, KF, RbF, CsF, LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, RbBr, CsBr, LiI, NaI, KI, RbI, CsI, and the like.
[0169] Examples of an alkaline earth metal halide may include BeF.sub.2, MgF.sub.2, CaF.sub.2, SrF.sub.2, BaF.sub.2, BeCl.sub.2, MgCl.sub.2, CaCl.sub.2, SrCl.sub.2, BaCl.sub.2, BeBr.sub.2, MgBr.sub.2, CaBr.sub.2, SrBr.sub.2, BaBr.sub.2, BeI.sub.2, MgI.sub.2, CaI.sub.2, SrI.sub.2, BaI.sub.2, and the like.
[0170] Examples of a transition metal halide may include a titanium halide (for example, TiF.sub.4, TiCl.sub.4, TiBr.sub.4, TiI.sub.4, etc.), a zirconium halide (for example, ZrF.sub.4, ZrCl.sub.4, ZrBr.sub.4, ZrI.sub.4, etc.), a hafnium halide (for example, HfF.sub.4, HfCl.sub.4, HfBr.sub.4, HfI.sub.4, etc.), a vanadium halide (for example, VF.sub.3, VCl.sub.3, VBr.sub.3, VI.sub.3, etc.), a niobium halide (for example, NbF.sub.3, NbCl.sub.3, NbBr.sub.3, NbI.sub.3, etc.), a tantalum halide (for example, TaF.sub.3, TaCl.sub.3, TaBr.sub.3, TaI.sub.3, etc.), a chromium halide (for example, CrF.sub.3, CrO.sub.3, CrBr.sub.3, CrI.sub.3, etc.), a molybdenum halide (for example, MoF.sub.3, MoCl.sub.3, MoBr.sub.3, MoI.sub.3, etc.), a tungsten halide (for example, WF.sub.3, WCl.sub.3, WBr.sub.3, WI.sub.3, etc.), a manganese halide (for example, MnF.sub.2, MnCl.sub.2, MnBr.sub.2, MnI.sub.2, etc.), a technetium halide (for example, TcF.sub.2, TcCl.sub.2, TcBr.sub.2, TcI.sub.2, etc.), a rhenium halide (for example, ReF.sub.2, ReCl.sub.2, ReBr.sub.2, ReI.sub.2, etc.), an iron halide (for example, FeF.sub.2, FeCl.sub.2, FeBr.sub.2, FeI.sub.2, etc.), a ruthenium halide (for example, RuF.sub.2, RuCl.sub.2, RuBr.sub.2, RuI.sub.2, etc.), an osmium halide (for example, OsF.sub.2, OSCl.sub.2, OsBr.sub.2, OSl.sub.2, etc.), a cobalt halide (for example, CoF.sub.2, COCl.sub.2, CoBr.sub.2, CoI.sub.2, etc.), a rhodium halide (for example, RhF.sub.2, RhCl.sub.2, RhBr.sub.2, RhI.sub.2, etc.), an iridium halide (for example, IrF.sub.2, IrCl.sub.2, IrBr.sub.2, IrI.sub.2, etc.), a nickel halide (for example, NiF.sub.2, NiCl.sub.2, NiBr.sub.2, NiI.sub.2, etc.), a palladium halide (for example, PdF.sub.2, PdCl.sub.2, PdBr.sub.2, PdI.sub.2, etc.), a platinum halide (for example, PtF.sub.2, PtCl.sub.2, PtBr.sub.2, PtI.sub.2, etc.), a copper halide (for example,

CuF, CuCl, CuBr, CuI, etc.), a silver halide (for example, AgF, AgCl, AgBr, AgI, etc.), a gold halide (for example, AuF, AuCl, AuBr, AuI, etc.), and the like.
[0171] Examples of a post-transition metal halide may include a zinc halide (for example, ZnF_2, $ZnCl_2$, $ZnBr_2$, ZnI_2, etc.), an indium halide (for example, InI_3, etc.), a tin halide (for example, SnI_2, etc.), and the like.
[0172] Examples of a lanthanide metal halide may include YbF, YbF_2, YbF_3, SmF_3, YbCl, $YbCl_2$, $YbCl_3$, $SmCl_3$, YbBr, $YbBr_2$, $YbBr_3$, $SmBr_3$, YbI, YbI_2, YbI_3, SmI_3, and the like.
[0173] Examples of a metalloid halide may include an antimony halide (for example, $SbCl_5$, etc.) and the like.
[0174] Examples of a metal telluride may include an alkali metal telluride (for example, Li_2Te, Na_2Te, K_2Te, Rb_2Te, Cs_2Te, etc.), an alkaline earth metal telluride (for example, BeTe, MgTe, CaTe, SrTe, BaTe, etc.), a transition metal telluride (for example, $TiTe_2$, $ZrTe_2$, $HfTe_2$, V_2Te_3, Nb_2Te_3, Ta_2Te_3, Cr_2Te_3, Mo_2Te_3, W_2Te_3, MnTe, TcTe, ReTe, FeTe, RuTe, OsTe, CoTe, RhTe, IrTe, NiTe, PdTe, PtTe, Cu_2Te, CuTe, Ag_2Te, AgTe, Au_2Te, etc.), a post-transition metal telluride (for example, ZnTe, etc.), a lanthanide metal telluride (for example, LaTe, CeTe, PrTe, NdTe, PmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, LuTe, etc.), and the like.
[Emission Layer **130**]
[0175] When the light-emitting device **10** is a full-color light-emitting device, the emission layer **130** may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer according to a subpixel. In embodiments, the emission layer **130** may have a stacked structure of two or more layers of a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers may contact each other or may be separated from each other to emit white light. In embodiments, the emission layer **130** may include two or more materials of a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials may be mixed with each other in a single layer to emit white light.
[0176] The emission layer **130** may include a host and a dopant. The dopant may include a phosphorescent dopant, a fluorescent dopant, or any combination thereof.
[0177] An amount of the dopant in the emission layer **130** may be in a range of about 0.01 parts by weight to about 15 parts by weight, based on 100 parts by weight of the host.
[0178] In embodiments, the emission layer **130** may include a quantum dot.
[0179] In embodiments, the emission layer **130** may include a delayed fluorescence material. The delayed fluorescence material may serve as a host or as a dopant in the emission layer **130**.
[0180] A thickness of the emission layer **130** may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the emission layer **130** may be in a range of about 200 Å to about 600 Å. When the thickness of the emission layer **130** is within any of these ranges, excellent luminescence characteristics may be obtained without a substantial increase in driving voltage.
[Host]
[0181] The host may include a compound represented by Formula 301:

$$[Ar_{301}]_{xb11}-[(L_{301})_{xb1}-R_{301}]_{xb21} \qquad \text{[Formula 301]}$$

[0182] In Formula 301, [0183] Ar_{301} and L_{301} may each independently be a C_3-C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C_1-C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a}, [0184] xb11 may be 1, 2, or 3, [0185] xb1 may be an integer from 0 to 5, [0186] R_{301} may be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C_1-C_{60} alkyl group unsubstituted or substituted with at least one R_{10a}, a C_2-C_{60} alkenyl group unsubstituted or substituted with at least one R_{10a}, a C_2-C_{60} alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(═O)(Q.sub.301), —S(═O).sub.2(Q.sub.301), or —P(═O)(Q.sub.301)(Q.sub.302), [0187] xb21 may be an integer from 1 to 5, and [0188] Q.sub.301 to Q.sub.303 may each independently be the same as described with respect to Q.sub.1.

[0189] In an embodiment, in Formula 301, when xb11 is 2 or more, two or more of Ar.sub.301(s) may be linked to each other via a single bond.

[0190] In an embodiment, the host may include a compound represented by Formula 301-1, a compound represented by Formula 301-2, or any combination thereof:

##STR00039##

[0191] In Formulae 301-1 and 301-2, [0192] ring A.sub.301 to ring A.sub.304 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0193] X.sub.301 may be O, S, N[(L.sub.304).sub.xb4-R.sub.304], C(R.sub.304)(R.sub.305), or Si(R.sub.304)(R.sub.305), [0194] xb22 and xb23 may each independently be 0, 1, or 2, [0195] L.sub.301, xb1, and R.sub.301 may each be the same as described herein, [0196] L.sub.302 to L.sub.304 may each independently be the same as described with respect to with L.sub.301, [0197] xb2 to xb4 may each independently be the same as described with respect to xb1, and [0198] R.sub.302 to R.sub.305 and R.sub.311 to R.sub.314 may each be the same as described with respect to R.sub.301.

[0199] In an embodiment, the host may include an alkali earth metal complex, a post-transition metal complex, or any combination thereof. For example, the host may include a Be complex (for example, Compound H55), an Mg complex, a Zn complex, or any combination thereof.

[0200] In an embodiment, the host may include one of Compounds H1 to H128, 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di-(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4′-bis(N-carbazolyl)-1,1′-biphenyl (CBP), 1,3-di(carbazol-9-yl)benzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), or any combination thereof:

##STR00040## ##STR00041## ##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047## ##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054## ##STR00055## ##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067##

[Phosphorescent Dopant]

[0201] The phosphorescent dopant may include at least one transition metal as a central metal.

[0202] The phosphorescent dopant may include a monodentate ligand, a bidentate ligand, a tridentate ligand, a tetradentate ligand, a pentadentate ligand, a hexadentate ligand, or any combination thereof.

[0203] The phosphorescent dopant may be electrically neutral.

[0204] In an embodiment, the phosphorescent dopant may include an organometallic compound represented by Formula 401:

##STR00068##

[0205] In Formulae 401 and 402, [0206] M may be a transition metal (for example, iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), gold (Au), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), rhenium (Re), or thulium (Tm)), [0207] L.sub.401 may be a ligand represented by Formula 402, and xc1 may be 1, 2, or 3, wherein when xc1 is two or more, two or more of L.sub.401(s) may be identical to or different from each other, [0208] L.sub.402 may

be an organic ligand, and xc2 may be 0, 1, 2, 3, or 4, wherein when xc2 is 2 or more, two or more of L.sub.402(s) may be identical to or different from each other, [0209] X.sub.401 and X.sub.402 may each independently be nitrogen or carbon, [0210] ring A.sub.401 and ring A.sub.402 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, [0211] T.sub.401 may be a single bond, *—O—*′, *—S—*′, *—C(═O)—*′, *—N(Q.sub.411)-*′, *—C(Q.sub.411)(Q.sub.412)-*′, *—C(Q.sub.411)=C(Q.sub.412)-*′, *—C(Q.sub.411)=*′, or *═C═*′, [0212] X.sub.403 and X.sub.404 may each independently be a chemical bond (for example, a covalent bond or a coordination bond), O, S, N(Q.sub.413), B(Q.sub.413), P(Q.sub.413), C(Q.sub.413)(Q.sub.414), or Si(Q.sub.413)(Q.sub.414), [0213] Q.sub.411 to Q.sub.414 may each independently be the same as described with respect to Q.sub.1, [0214] R.sub.401 and R.sub.402 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.20 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(═O)(Q.sub.401), —S(═O).sub.2(Q.sub.401), or —P(═O)(Q.sub.401)(Q.sub.402), [0215] Q.sub.401 to Q.sub.403 may each independently be the same as described with respect to Q.sub.1, [0216] xc11 and xc12 may each independently be an integer from 0 to 10, and [0217] * and *′ in Formula 402 each indicate a binding site to M in Formula 401.
[0218] For example, in Formula 402, X.sub.401 may be nitrogen and X.sub.402 may be carbon, or X.sub.401 and X.sub.402 may each be nitrogen.
[0219] In an embodiment, in Formula 401, when xc1 is 2 or more, two ring A.sub.401(s) among two or more of L.sub.401 may optionally be linked to each other via T.sub.402, which is a linking group, and two ring A.sub.402(s) among two or more of L.sub.401 may optionally be linked to each other via T.sub.403, which is a linking group (see Compounds PD1 to PD4 and PD7). T.sub.402 and T.sub.403 may each independently be the same as described with respect to T.sub.401.
[0220] In Formula 401, L.sub.402 may be an organic ligand. For example, L.sub.402 may include a halogen group, a diketone group (for example, an acetylacetonate group), a carboxylic acid group (for example, a picolinate group), —C(═O), an isonitrile group, —CN group, a phosphorus group (for example, a phosphine group, a phosphite group, etc.), or any combination thereof.
[0221] In an embodiment, the phosphorescent dopant may include, for example, one of Compounds PD1 to PD39, or any combination thereof:
##STR00069## ##STR00070## ##STR00071## ##STR00072## ##STR00073## ##STR00074## ##STR00075##
[Fluorescent Dopant]
[0222] The fluorescent dopant may include an amine group-containing compound, a styryl group-containing compound, or any combination thereof.
[0223] In an embodiment, the fluorescent dopant may include a compound represented by Formula 501:
##STR00076##
[0224] In Formula 501, [0225] Ar.sub.501, L.sub.501 to L.sub.503, R.sub.501, and R.sub.502 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0226] xd1 to xd3 may each independently be 0, 1, 2, or 3, and [0227] xd4 may be 1, 2, 3, 4, 5, or 6.
[0228] In an embodiment, in Formula 501, Ar.sub.501 may be a condensed cyclic group (for example, an anthracene group, a chrysene group, a pyrene group, etc.) in which three or more

monocyclic groups are condensed together.
[0229] In an embodiment, in Formula 501, xd4 in may be 2.
[0230] In an embodiment, the fluorescent dopant may include one of Compounds FD1 to FD37, DPVBi, DPAVBi, or any combination thereof:
##STR00077## ##STR00078## ##STR00079## ##STR00080## ##STR00081## ##STR00082## ##STR00083##
[Delayed Fluorescence Material]
[0231] The emission layer **130** may include a delayed fluorescence material.
[0232] In the specification, a delayed fluorescence material may be any compound that is capable of emitting delayed fluorescence, based on a delayed fluorescence emission mechanism.
[0233] The delayed fluorescence material included in the emission layer **130** may serve as a host or a as dopant, depending on the types of other materials included in the emission layer **130**.
[0234] According to an embodiment, a difference between a triplet energy level (eV) of the delayed fluorescence material and a singlet energy level (eV) of the delayed fluorescence material may be in a range of about 0 eV to about 0.5 eV. When the difference between the triplet energy level (eV) of the delayed fluorescence material and the singlet energy level (eV) of the delayed fluorescence material is within the range above, up-conversion from the triplet state to the singlet state of the delayed fluorescence materials may effectively occur, and thus, the light-emitting device **10** may have improved luminescence efficiency.
[0235] In an embodiment, the delayed fluorescence material may include: a material including at least one electron donor (for example, a π electron-rich C.sub.3-C.sub.60 cyclic group, such as a carbazole group) and at least one electron acceptor (for example, a sulfoxide group, a cyano group, or a π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group); a material including a C.sub.8-C.sub.60 polycyclic group in which two or more cyclic groups are condensed while sharing a boron (B) atom; or the like.
[0236] In an embodiment, the delayed fluorescence material may include at least one of Compounds DF1 to DF14:
##STR00084## ##STR00085## ##STR00086## ##STR00087##
[Quantum Dot]
[0237] The emission layer **130** may include a quantum dot.
[0238] In the specification, a quantum dot may be a crystal of a semiconductor compound. Quantum dots may emit light of various emission wavelengths depending on a size of the crystal. Quantum dots may also emit light of various emission wavelengths by adjusting a ratio of elements constituting the quantum dots.
[0239] A diameter of a quantum dot may be, for example, in a range of about 1 nm to about 10 nm.
[0240] The quantum dot may be synthesized by a wet chemical process, a metal organic chemical vapor deposition process, a molecular beam epitaxy process, or any process similar thereto.
[0241] The wet chemical process is a method that includes mixing a precursor material with an organic solvent and growing a quantum dot particle crystal. When the crystal grows, the organic solvent naturally serves as a dispersant coordinated on the surface of the quantum dot crystal and controls the growth of the crystal so that the growth of quantum dot particles may be controlled through a process which costs less, and may be more readily performed than vapor deposition methods, such as metal organic chemical vapor deposition (MOCVD) or molecular beam epitaxy (MBE).
[0242] A quantum dot may include a Group II-VI semiconductor compound, a Group III-V semiconductor compound, a Group III-VI semiconductor compound, a Group I-III-VI semiconductor compound, a Group IV-VI semiconductor compound, a Group IV element or compound, or any combination thereof.
[0243] Examples of a Group II-VI semiconductor compound may include: a binary compound, such as CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnO, HgS, HgSe, HgTe, MgSe, MgS, etc.; a ternary

compound, such as CdSeS, CdSeTe, CdSTe, ZnSeS, ZnSeTe, ZnSTe, HgSeS, HgSeTe, HgSTe, CdZnS, CdZnSe, CdZnTe, CdHgS, CdHgSe, CdHgTe, HgZnS, HgZnSe, HgZnTe, MgZnSe, MgZnS, etc.; a quaternary compound, such as CdZnSeS, CdZnSeTe, CdZnSTe, CdHgSeS, CdHgSeTe, CdHgSTe, HgZnSeS, HgZnSeTe, HgZnSTe, etc.; and any combination thereof.
[0244] Examples of a Group III-V semiconductor compound may include: a binary compound, such as GaN, GaP, GaAs, GaSb, AlN, AlP, AlAs, AlSb, InN, InP, InAs, InSb, etc.; a ternary compound, such as GaNP, GaNAs, GaNSb, GaPAs, GaPSb, AlNP, AlNAs, AlNSb, AlPAs, AlPSb, InGaP, InNP, InAlP, InNAs, InNSb, InPAs, InPSb, etc.; a quaternary compound, such as GaAlNP, GaAlNAs, GaAlNSb, GaAlPAs, GaAlPSb, GaInNP, GaInNAs, GaInNSb, GaInPAs, GaInPSb, InAlNP, InAlNAs, InAlNSb, InAlPAs, InAlPSb, etc.; and any combination thereof. In embodiments, a Group III-V semiconductor compound may further include a Group II element. Examples of a Group III-V semiconductor compound further including a Group II element may include InZnP, InGaZnP, InAlZnP, etc.
[0245] Examples of a Group III-VI semiconductor compound may include: a binary compound, such as GaS, GaSe, Ga.sub.2Se.sub.3, GaTe, InS, InSe, In.sub.2S.sub.3, In.sub.2Se.sub.3, InTe, etc.; a ternary compound, such as InGaS.sub.3, InGaSe.sub.3, etc.; and any combination thereof.
[0246] Examples of a Group I-III-VI semiconductor compound may include: a ternary compound, such as AgInS, AgInS.sub.2, AgInSe.sub.2, AgGaS, AgGaS.sub.2, AgGaSe.sub.2, CuInS, CuInS.sub.2, CuInSe.sub.2, CuGaS.sub.2, CuGaSe.sub.2, CuGaO.sub.2, AgGaO.sub.2, AgAlO.sub.2, etc.; a quaternary compound, such as AgInGaS.sub.2, AgInGaSe.sub.2, etc.; and any combination thereof.
[0247] Examples of a Group IV-VI semiconductor compound may include: a binary compound, such as SnS, SnSe, SnTe, PbS, PbSe, PbTe, etc.; a ternary compound, such as SnSeS, SnSeTe, SnSTe, PbSeS, PbSeTe, PbSTe, SnPbS, SnPbSe, SnPbTe, etc.; a quaternary compound, such as SnPbSSe, SnPbSeTe, SnPbSTe, etc.; and any combination thereof.
[0248] Examples of a Group IV element or compound may include: a single element, such as Si, Ge, etc.; a binary compound, such as SiC, SiGe, etc.; and any combination thereof.
[0249] Each element included in a compound, such as a binary compound, a ternary compound, or a quaternary compound, may be present in a particle at a uniform concentration or at a non-uniform concentration. A formula of a compound as listed above may indicate the elements that are included in the compound, wherein the ratios of the constituent elements in the compound may vary. For example, AgInGaS.sub.2 may refer to AgIn.sub.xGa.sub.1-xS.sub.2 (where x is a real number between 0 and 1).
[0250] In embodiments, a quantum dot may have a single structure in which the concentration of each element in the quantum dot is uniform, or a quantum dot may have a core-shell structure. For example, when a quantum dot has a core-shell structure, a material included in the core and a material included in the shell may be different from each other.
[0251] The shell of a quantum dot may serve as a protective layer that prevents chemical degeneration of the core to maintain semiconductor characteristics, and/or may serve as a charging layer that imparts electrophoretic characteristics to the quantum dot. The shell may be single-layered or multilayered. An interface between the core and the shell may have a concentration gradient in which the concentration of an element that is present in the shell decreases toward the core.
[0252] A shell of a quantum dot may include a metal oxide, a non-metal oxide, a semiconductor compound, or any combination thereof. Examples of a metal oxide or a non-metal oxide may include: a binary compound, such as SiO.sub.2, Al.sub.2O.sub.3, TiO.sub.2, ZnO, MnO, Mn.sub.2O.sub.3, Mn.sub.3O.sub.4, CuO, FeO, Fe.sub.2O.sub.3, Fe.sub.3O.sub.4, CoO, Co.sub.3O.sub.4, NiO, etc.; a ternary compound, such as MgAl.sub.2O.sub.4, CoFe.sub.2O.sub.4, NiFe.sub.2O.sub.4, CoMn.sub.2O.sub.4, etc.; and any combination thereof.
[0253] Examples of a semiconductor compound may include, as described herein: a Group III-VI

semiconductor compound; a Group II-VI semiconductor compound; a Group III-V semiconductor compound; a Group III-VI semiconductor compound; a Group I-III-VI semiconductor compound; a Group IV-VI semiconductor compound; and any combination thereof. For example, the semiconductor compound may include CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnSeS, ZnTeS, GaAs, GaP, GaS, GaSe, AgGaS, AgGaS.sub.2, GaSb, HgS, HgSe, HgTe, InAs, InP, InGaP, InSb, AlAs, AlP, AlSb, or any combination thereof.

[0254] The quantum dot may have a full width at half maximum (FWHM) of an emission spectrum equal to or less than about 45 nm. For example, the quantum dot may have a FWHM of an emission spectrum equal to or less than about 40 nm. For example, the quantum dot may have an FWHM of an emission spectrum equal to or less than about 30 nm. When the FWHM of the quantum dot is within any of these ranges, the quantum dot may have improved color purity or improved color reproducibility. Light emitted through the quantum dot may be emitted in all directions, so that a wide viewing angle may be improved.

[0255] In embodiments, a quantum dot may be in the form of a spherical particle, a pyramidal particle, a multi-arm particle, a cubic nanoparticles, a nanotube particle, a nanowire particle, a nanofiber particle, a nanoplate particle, or the like.

[0256] By adjusting the size of the quantum dots or the ratio of elements in a quantum dot compound, the energy band gap may be adjusted, and thus, light of various wavelengths may be obtained from a quantum dot emission layer. Therefore, by using the aforementioned quantum dots (using quantum dots of different sizes or having different element ratios in the quantum dot compound), a light-emitting device emitting light of various wavelengths may be implemented. In embodiments, the size of the quantum dots or the ratio of elements in the quantum dot compound may be selected to emit red light, green light, and/or blue light. In embodiments, the quantum dots may be configured to emit white light by a combination of light of various colors.

[Electron Transport Region **140**]

[0257] The electron transport region **140** may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer including different materials, or a structure including multiple layers including different materials.

[0258] The electron transport region **140** may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

[0259] In embodiments, the electron transport region **140** may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein the layers of each structure may be stacked from the emission layer **130** in its respective stated order, but the structure of the electron transport region **140** is not limited thereto.

[0260] The electron transport region **140** (for example, a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region **140**) may include a metal-free compound including at least one π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group.

[0261] In an embodiment, the electron transport region **140** may include a compound represented by Formula 601:

[Ar.sub.601].sub.xe11-[(L.sub.611).sub.xe1-R.sub.601].sub.xe21 [Formula 601]

[0262] In Formula 601, [0263] Ar.sub.601 and L.sub.601 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0264] xe11 may be 1, 2, or 3, [0265] xe1 may be 0, 1, 2, 3, 4, or 5, [0266] R.sub.601 may be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-

C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.601)(Q.sub.602)(Q.sub.603), —C(═O)(Q.sub.601), —S(═O).sub.2(Q.sub.601), or —P(═O)(Q.sub.601)(Q.sub.602), [0267] Q.sub.601 to Q.sub.603 may each independently be the same as described with respect to Q.sub.1, [0268] xe21 may be 1, 2, 3, 4, or 5, and [0269] at least one of Ar.sub.601, L.sub.601, and R.sub.601 may each independently be a π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a.
[0270] In an embodiment, in Formula 601, when xe11 is 2 or more, two or more of Ar.sub.601 may be linked to each other via a single bond.
[0271] In an embodiment, in Formula 601, Ar.sub.601 may be an anthracene group unsubstituted or substituted with at least one R.sub.10a.
[0272] In an embodiment, the electron transport region **140** may include a compound represented by Formula 601-1:
##STR00088##
[0273] In Formula 601-1, [0274] X.sub.614 may be N or C(R.sub.614), X.sub.615 may be N or C(R.sub.615), X.sub.616 may be N or C(R.sub.616), and at least one of X.sub.614 to X.sub.616 may each be N, [0275] L.sub.611 to L.sub.613 may each independently be the same as described with respect to L.sub.601, [0276] xe611 to xe613 may each independently be the same as described with respect to xe1, [0277] R.sub.611 to R.sub.613 may each independently be the same as described with respect to R.sub.601, and [0278] R.sub.614 to R.sub.616 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.
[0279] In an embodiment, in Formulae 601 and 601-1, xe1 and xe611 to xe613 may each independently be 0, 1, or 2.
[0280] In an embodiment, the electron transport region **140** may include one of Compounds ET1 to ET45, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq.sub.3, BAlq, TAZ, NTAZ, or any combination thereof:
##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103##
[0281] A thickness of the electron transport region **140** may be in a range of about 100 Å to about 5,000 Å. For example, the thickness of the electron transport region **140** may be in a range of about 160 Å to about 4,000 Å. When the electron transport region **140** includes a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, or any combination thereof, a thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be in a range of about 20 Å to about 1,000 Å, and a thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be in a range of about 30 Å to about 300 Å. For example, the thickness of the electron transport layer may be in a range of about 150 Å to about 500 Å. When the thicknesses of the buffer layer, the hole blocking layer, the electron control layer, the electron transport layer, and/or the electron transport region **140** are within these ranges, satisfactory electron transporting characteristics may be obtained without a substantial increase in driving voltage.
[0282] The electron transport region **140** (for example, an electron transport layer in the electron transport region **140**) may further include, in addition to the aforementioned materials, a metal-containing material.
[0283] The metal-containing material may include an alkali metal complex, an alkaline earth metal complex, or any combination thereof. A metal ion of an alkali metal complex may be a Li ion, a Na

ion, a K ion, a Rb ion, or a Cs ion; and a metal ion of an alkaline earth metal complex may be a Be ion, a Mg ion, a Ca ion, a Sr ion, or a Ba ion.
[0284] A ligand coordinated with a metal ion of an alkali metal complex or an alkaline earth metal complex may each independently include a hydroxyquinoline, a hydroxyisoquinoline, a hydroxybenzoquinoline, a hydroxyacridine, a hydroxyphenanthridine, a hydroxyphenyloxazole, a hydroxyphenylthiazole, a hydroxyphenyloxadiazole, a hydroxyphenylthiadiazole, a hydroxyphenylpyridine, a hydroxyphenylbenzimidazole, a hydroxyphenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.
[0285] In an embodiment, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (Liq) or Compound ET-D2:
##STR00104##
[0286] The electron transport region **140** may include an electron injection layer that facilitates the injection of electrons from the second electrode **150**. The electron injection layer may directly contact the second electrode **150**.
[0287] The electron injection layer may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer including different materials, or a structure including multiple layers including different materials.
[0288] The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof.
[0289] The alkali metal may include Li, Na, K, Rb, Cs, or any combination thereof. The alkaline earth metal may include Mg, Ca, Sr, Ba, or any combination thereof. The rare earth metal may include Sc, Y, Ce, Tb, Yb, Gd, or any combination thereof.
[0290] The alkali metal-containing compound, the alkaline earth metal-containing compound, and the rare earth metal-containing compound may include oxides, halides (for example, fluorides, chlorides, bromides, iodides, etc.), or tellurides of the alkali metal, the alkaline earth metal, and the rare earth metal, or any combination thereof.
[0291] The alkali metal-containing compound may include: an alkali metal oxide, such as Li.sub.2O, Cs.sub.2O, K.sub.2O, etc.; an alkali metal halide, such as LiF, NaF, CsF, KF, LiI, NaI, CsI, KI, etc.; or any combination thereof. The alkaline earth metal-containing compound may include an alkaline earth metal oxide, such as BaO, SrO, CaO, Ba.sub.xSr.sub.1-xO (wherein x is a real number satisfying the condition of 0<x<1), Ba.sub.xCa.sub.1-xO (wherein x is a real number satisfying the condition of 0<x<1), etc. The rare earth metal-containing compound may include YbF.sub.3, ScF.sub.3, Sc.sub.2O.sub.3, Y.sub.2O.sub.3, Ce.sub.2O.sub.3, GdF.sub.3, TbF.sub.3, YbI.sub.3, ScI.sub.3, TbI.sub.3, or any combination thereof. In embodiments, the rare earth metal-containing compound may include a lanthanide metal telluride. Examples of a lanthanide metal telluride may include LaTe, CeTe, PrTe, NdTe, PmTe, SmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, LuTe, La.sub.2Te.sub.3, Ce.sub.2Te.sub.3, Pr.sub.2Te.sub.3, Nd.sub.2Te.sub.3, Pm.sub.2Te.sub.3, Sm.sub.2Te.sub.3, Eu.sub.2Te.sub.3, Gd.sub.2Te.sub.3, Tb.sub.2Te.sub.3, Dy.sub.2Te.sub.3, Ho.sub.2Te.sub.3, Er.sub.2Te.sub.3, Tm.sub.2Te.sub.3, Yb.sub.2Te.sub.3, Lu.sub.2Te.sub.3, etc.
[0292] The alkali metal complex, the alkaline earth metal complex, and the rare earth metal complex may include: an alkali metal ion, an alkaline earth metal ion, or a rare earth metal ion; and a ligand bonded to the metal ion (for example, hydroxyquinoline, hydroxyisoquinoline, hydroxybenzoquinoline, hydroxyacridine, hydroxyphenanthridine, hydroxyphenyloxazole, hydroxyphenylthiazole, hydroxyphenyloxadiazole, hydroxyphenylthiadiazole, hydroxyphenylpyridine, hydroxyphenylbenzimidazole, hydroxyphenylbenzothiazole, bipyridine, phenanthroline, cyclopentadiene, or any combination thereof).
[0293] In an embodiment, the electron injection layer may consist of an alkali metal, an alkaline

earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof, as described above. In embodiments, the electron injection layer may further include an organic material (for example, a compound represented by Formula 601).

[0294] According to an embodiment, the electron injection layer may consist of an alkali metal-containing compound (for example, an alkali metal halide); or the electron injection layer may consist of an alkali metal-containing compound (for example, an alkali metal halide), and an alkali metal, an alkaline earth metal, a rare earth metal, or any combination thereof. For example, the electron injection layer may be a Kl:Yb co-deposited layer, an RbI:Yb co-deposited layer, a LiF:Yb co-deposited layer, or the like.

[0295] When the electron injection layer further includes an organic material, the alkali metal, the alkaline earth metal, the rare earth metal, the alkali metal-containing compound, the alkaline earth metal-containing compound, the rare earth metal-containing compound, the alkali metal complex, the alkaline earth metal complex, the rare earth metal complex, or any combination thereof may be uniformly or non-uniformly dispersed in a matrix including the organic material.

[0296] A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å. For example, the thickness of the electron injection layer may be in a range of about 3 Å to about 90 Å. When the thickness of the electron injection layer is within any of the ranges described above, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

[Second Electrode **150**]

[0297] The second electrode **150** may be arranged on the electron transport region **140**. The second electrode **150** may be a cathode, which is an electron injection electrode. When the second electrode **150** is a cathode, the second electrode **150** may include a material having a low-work function, such as a metal, an alloy, an electrically conductive compound, or any combination thereof.

[0298] The second electrode **150** may include Li, Ag, Mg, Al, Al—Li, Ca, Mg—In, Mg—Ag, Yb, Ag—Yb, ITO, IZO, or any combination thereof. The second electrode **150** may be a transmissive electrode, a transflective electrode, or a reflective electrode.

[0299] The second electrode **150** may have a single-layered structure or a multi-layered structure.

[Capping Layer]

[0300] The light-emitting device **10** may include a first capping layer arranged outside the first electrode **110**, and/or a second capping layer arranged outside the second electrode **150**. In embodiments, the light-emitting device **10** may have a structure in which the first capping layer, the first electrode **110**, the interlayer, and the second electrode **150** are stacked in the stated order, a structure in which the first electrode **110**, the interlayer, the second electrode **150**, and the second capping layer are stacked in the stated order, or a structure in which the first capping layer, the first electrode **110**, the interlayer, the second electrode **150**, and the second capping layer are stacked in the stated order.

[0301] Light generated in the emission layer **130** of the light-emitting device **10** may pass through the first electrode **110**, which may be a transflective electrode or a transmissive electrode, and through the first capping layer to the outside. Light generated in the emission layer **130** of the light-emitting device **10** may pass through the second electrode **150**, which may be a transflective electrode or a transmissive electrode, and through the second capping layer to the outside.

[0302] The first capping layer and the second capping layer may each increase external luminescence efficiency according to the principle of constructive interference. Accordingly, light extraction efficiency of the light-emitting device **10** is increased, so that the luminescence efficiency of the light-emitting device **10** may be improved.

[0303] The first capping layer and the second capping layer may each include a material having a

refractive index equal to or greater than about 1.2 (with respect to a wavelength of about 460 nm).
[0304] The first capping layer and the second capping layer may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or an organic-inorganic composite capping layer including an organic material and an inorganic material.
[0305] At least one of the first capping layer and the second capping layer may each independently include a carbocyclic compound, a heterocyclic compound, an amine group-containing compound, a porphine derivative, a phthalocyanine derivative, a naphthalocyanine derivative, an alkali metal complex, an alkaline earth metal complex, or any combination thereof. The carbocyclic compound, the heterocyclic compound, and the amine group-containing compound may optionally be substituted with a substituent including O, N, S, Se, Si, F, Cl, Br, I, or any combination thereof.
[0306] According to an embodiment, at least one of the first capping layer and the second capping layer may each independently include an amine group-containing compound.
[0307] For example, at least one of the first capping layer and the second capping layer may each independently include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof.
[0308] In an embodiment, at least one of the first capping layer and the second capping layer may each independently include one of Compounds HT28 to HT33, one of Compounds CP1 to CP6, β-NPB, or any combination thereof:
##STR00105## ##STR00106##
[Film]
[0309] The electronic apparatus, which will be further described below, may further include a film. The film may be, for example, an optical member (or a light control means) (for example, a color filter, a color conversion member, a capping layer, a light extraction efficiency enhancement layer, a selective light absorbing layer, a polarizing layer, a quantum dot-containing layer, etc.), a light blocking member (for example, a light reflective layer, a light absorbing layer, etc.), a protective member (for example, an insulating layer, a dielectric layer, etc.), or the like.
[Electronic Apparatus]
[0310] The light-emitting device **10** may be included in various electronic apparatuses. For example, an electronic apparatus including the light-emitting device **10** may be a display apparatus, an authentication apparatus, or the like.
[0311] The electronic apparatus (for example, a display apparatus) may further include, in addition to the light-emitting device **10**, a color filter, a color conversion layer, or a color filter and a color conversion layer. The color filter and/or the color conversion layer may be arranged in at least one direction in which light emitted from the light-emitting device **10** travels. For example, the light emitted from the light-emitting device **10** may be blue light or white light. Further details on the light-emitting device **10** may be the same as described herein. According to an embodiment, the color conversion layer may include a quantum dot. The quantum dot may be, for example, a quantum dot as described herein.
[0312] The electronic apparatus may include a substrate. The substrate may include subpixels, the color filter may include color filter areas respectively corresponding to the subpixels, and the color conversion layer may include color conversion areas respectively corresponding to the subpixels.
[0313] A pixel defining layer may be arranged between the subpixels to define each subpixel.
[0314] The color filter may further include color filter areas and light-shielding patterns arranged between the color filter areas, and the color conversion layer may further include color conversion areas and light-shielding patterns arranged between the color conversion areas.
[0315] The color filter areas (or the color conversion areas) may include a first area emitting first color light, a second area emitting second color light, and/or a third area emitting third color light, wherein the first color light, the second color light, and/or the third color light may have different maximum emission wavelengths. For example, the first color light may be red light, the second

color light may be green light, and the third color light may be blue light. In an embodiment, the color filter areas (or the color conversion areas) may include quantum dots. For example, the first area may include a red quantum dot, the second area may include a green quantum dot, and the third area may not include a quantum dot. The quantum dots may be quantum dots as described herein. The first area, the second area, and/or the third area may each further include a scatterer.
[0316] In an embodiment, the light-emitting device **10** may emit first light, the first area may absorb the first light to emit first-first color light, the second area may absorb the first light to emit second-first color light, and the third area may absorb the first light to emit third-first color light. The first-first color light, the second-first color light, and the third-first color light may have different maximum emission wavelengths. For example, the first light may be blue light, the first-first color light may be red light, the second-first color light may be green light, and the third-first color light may be blue light.
[0317] The electronic apparatus may further include a thin-film transistor, in addition to the light-emitting device **10** as described above. The thin-film transistor may include a source electrode, a drain electrode, and an active layer, wherein any one of the source electrode and the drain electrode may be electrically connected to any one of the first electrode and the second electrode of the light-emitting device.
[0318] The thin-film transistor may further include a gate electrode, a gate insulating film, or the like.
[0319] The active layer may include crystalline silicon, amorphous silicon, an organic semiconductor, an oxide semiconductor, or the like.
[0320] The electronic apparatus may further include a sealing portion for sealing the light-emitting device. The sealing portion may be arranged between the color filter and/or the color conversion layer and the light-emitting device. The sealing portion allows light from the light-emitting device **10** to be extracted to the outside, and simultaneously prevents ambient air and moisture from penetrating into the light-emitting device **10**. The sealing portion may be a sealing substrate including a transparent glass substrate or a plastic substrate. The sealing portion may be a thin-film encapsulation layer that includes at least one of an organic layer and/or an inorganic layer. When the sealing portion is a thin-film encapsulation layer, the electronic apparatus may be flexible.
[0321] Various functional layers may be further included on the sealing portion, in addition to the color filter and/or the color conversion layer, according to a use of the electronic apparatus. Examples of a functional layer may include a touch screen layer, a polarizing layer, and the like. The touch screen layer may be a pressure-sensitive touch screen layer, a capacitive touch screen layer, or an infrared touch screen layer. The authentication apparatus may be, for example, a biometric authentication apparatus that authenticates an individual by using biometric information of a living body (for example, fingertips, pupils, etc.).
[0322] The authentication apparatus may further include, in addition to the light-emitting device **10** as described above, a biometric information collector.
[0323] The electronic apparatus may be applied to various displays, light sources, lighting, personal computers (for example, a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (for example, electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram displays, ultrasonic diagnostic devices, or endoscope displays), fish finders, various measuring instruments, meters (for example, meters for a vehicle, an aircraft, and a vessel), projectors, and the like.
[Electronic Equipment]
[0324] The light-emitting device **10** may be included in various electronic equipment.
[0325] In embodiments, an electronic equipment including the light-emitting device **10** may be a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent

display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.

[0326] Since the light-emitting device **10** has improved luminescence efficiency and improved lifespan, the electronic equipment including the light-emitting device **10** may have characteristics such as high luminance, high resolution, and low power consumption.

Descriptions of FIGS. **3** and **4**

[0327] FIG. **3** is a schematic cross-sectional view of an electronic apparatus according to an embodiment.

[0328] The electronic apparatus of FIG. **3** may include a substrate **100**, a thin-film transistor (TFT), a light-emitting device, and an encapsulation portion **300**.

[0329] The substrate **100** may be a flexible substrate, a glass substrate, or a metal substrate. A buffer layer **210** may be arranged on the substrate **100**. The buffer layer **210** may prevent penetration of impurities through the substrate **100** and may provide a flat surface on the substrate **100**.

[0330] A TFT may be arranged on the buffer layer **210**. The TFT may include an active layer **220**, a gate electrode **240**, a source electrode **260**, and a drain electrode **270**.

[0331] The active layer **220** may include an inorganic semiconductor such as silicon or polysilicon, an organic semiconductor, or an oxide semiconductor, and may include a source region, a drain region, and a channel region.

[0332] A gate insulating film **230** for insulating the active layer **220** from the gate electrode **240** may be arranged on the active layer **220**, and the gate electrode **240** may be arranged on the gate insulating film **230**.

[0333] An interlayer insulating film **250** may be arranged on the gate electrode **240**. The interlayer insulating film **250** may be arranged between the gate electrode **240** and the source electrode **260** to insulate the gate electrode **240** from the source electrode **260** and between the gate electrode **240** and the drain electrode **270** to insulate the gate electrode **240** from the drain electrode **270**.

[0334] The source electrode **260** and the drain electrode **270** may be arranged on the interlayer insulating film **250**. The interlayer insulating film **250** and the gate insulating film **230** may be formed to expose a source region and a drain region of the active layer **220**, and the source electrode **260** and the drain electrode **270** may respectively contact the exposed portions of the source region and the drain region of the active layer **220**.

[0335] The TFT may be electrically connected to a light-emitting device to drive the light-emitting device, and may be covered and protected by a passivation layer **280**. The passivation layer **280** may include an inorganic insulating film, an organic insulating film, or any combination thereof. The light-emitting device may be provided on the passivation layer **280**. The light-emitting device may include the first electrode **110**, the interlayer, and the second electrode **150**.

[0336] The first electrode **110** may be arranged on the passivation layer **280**. The passivation layer **280** may not completely cover the drain electrode **270** and may expose a portion of the drain electrode **270**. The first electrode **110** may be connected (for example, electrically connected) to the exposed portion of the drain electrode **270**.

[0337] A pixel defining layer **290** including an insulating material may be arranged on the first electrode **110**. The pixel defining layer **290** may expose a portion of the first electrode **110**, and the interlayer may be formed on the exposed portion of the first electrode **110**. The pixel defining layer **290** may be a polyimide or polyacrylic organic film. Although not shown in FIG. **3**, at least some layers of the interlayer may extend beyond the upper portion of the pixel defining layer **290** to be provided in the form of a common layer.

[0338] The second electrode **150** may be arranged on the interlayer, and a capping layer **170** may be further included on the second electrode **150**. The capping layer **170** may be formed to cover the second electrode **150**.
[0339] The encapsulation portion **300** may be arranged on the capping layer **170**. The encapsulation portion **300** may be arranged on a light-emitting device to protect the light-emitting device from moisture and/or oxygen. The encapsulation portion **300** may include: an inorganic film including silicon nitride (SiN.sub.x), silicon oxide (SiO.sub.x), indium tin oxide, indium zinc oxide, or any combination thereof; an organic film including polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyimide, polyethylene sulfonate, polyoxymethylene, polyarylate, hexamethyldisiloxane, an acrylic resin (for example, polymethyl methacrylate, polyacrylic acid, etc.), an epoxy-based resin (for example, aliphatic glycidyl ether (AGE), etc.), or any combination thereof; or any combination of the inorganic film and the organic film.
[0340] FIG. **4** is a schematic cross-sectional view of an electronic apparatus according to another embodiment.
[0341] The electronic apparatus of FIG. **4** may differ from the electronic apparatus of FIG. **3**, at least in that a light-shielding pattern **500** and a functional region **400** are further included on the encapsulation portion **300**. The functional region **400** may be a color filter area, a color conversion area, or a combination of the color filter area and the color conversion area. According to an embodiment, the light-emitting device included in the electronic apparatus of FIG. **4** may be a tandem light-emitting device.
Description of FIG. **5**
[0342] FIG. **5** is a schematic perspective view of an electronic equipment **1** including a light-emitting device according to an embodiment.
[0343] The electronic equipment **1**, which may be an apparatus that displays a moving image or a still image, may be not only a portable electronic equipment, such as a mobile phone, a smartphone, a tablet computer, a mobile communication terminal, an electronic notebook, an electronic book, a portable multimedia player (PMP), a navigation device, or an ultra-mobile PC (UMPC), but may also be various products, such as a television, a laptop computer, a monitor, a billboard, or an Internet of things (IOT). The electronic equipment **1** may be any product as described above or a part thereof.
[0344] In an embodiment, the electronic equipment **1** may be a wearable device, such as a smart watch, a watch phone, a glasses-type display, or a head mounted display (HMD), or a part of the wearable device. However, embodiments are not limited thereto.
[0345] Examples of the electronic equipment **1** may include a dashboard of a vehicle, a center information display on a center fascia or dashboard of a vehicle, a room mirror display that replaces a side mirror of a vehicle, an entertainment display arranged for a rear seat of a vehicle or arranged on the back of a front seat, a head-up display (HUD) installed at the front of a vehicle or projected onto a front window glass, or a computer generated hologram augmented reality head up display (CGH AR HUD). FIG. **5** illustrates an embodiment in which the electronic equipment **1** is a smartphone, for convenience of explanation.
[0346] The electronic equipment **1** may include a display area DA and a non-display area NDA outside the display area DA. The electronic equipment **1** may implement an image through a two-dimensional array of pixels that are arranged in the display area DA.
[0347] The non-display area NDA is an area that does not display an image, and may surround (for example, entirely surround) the display area DA. A driver for providing electrical signals or power to display devices arranged on the display area DA may be arranged in the non-display area NDA. A pad, which is an area to which an electronic element or a printed circuit board may be electrically connected, may be arranged in the non-display area NDA.
[0348] In the electronic equipment **1**, a length in the x-axis direction and a length in the y-axis direction may be different from each other. For example, as shown in FIG. **5**, the length in the x-

axis direction may be less than the length in the y-axis direction. As another example, the length in the x-axis direction may be the same as the length in the y-axis direction. As another example, the length in the x-axis direction may be longer than the length in the y-axis direction.

Descriptions of FIGS. **6** and **7**A to **7**C

[0349] FIG. **6** is a schematic perspective view of an exterior of a vehicle **1000** as an electronic equipment including a light-emitting device according to an embodiment. FIGS. **7**A to **7**C are each a schematic diagram of an interior of a vehicle **1000** according to embodiments.

[0350] Referring to FIGS. **6** and **7**A to **7**C, embodiments of a vehicle **1000** may include various apparatuses for moving a subject to be transported, such as a person, an object, or an animal, from a departure point to a destination. Examples of a vehicle **1000** may include a vehicle traveling on a road or track, a vessel moving over a sea or river, an airplane flying in the sky using the action of air, and the like.

[0351] The vehicle **1000** may travel on a road or a track. The vehicle **1000** may move in a selected or given direction according to the rotation of at least one wheel. Examples of a vehicle **1000** may include a three-wheeled or four-wheeled vehicle, a construction machine, a two-wheeled vehicle, a prime mover device, a bicycle, and a train running on a track.

[0352] The vehicle **1000** may include a vehicle body having an interior and an exterior, and a chassis that is a portion excluding the vehicle body in which mechanical apparatuses necessary for driving are installed. The exterior of the vehicle body may include a front panel, a bonnet, a roof panel, a rear panel, a trunk, a pillar provided at a boundary between doors, and the like. The chassis of the vehicle **1000** may include a power generating device, a power transmitting device, a driving device, a steering device, a braking device, a suspension device, a transmission device, a fuel device, front and rear wheels, left and right wheels, and the like.

[0353] The vehicle **1000** may include a side window glass **1100**, a front window glass **1200**, a side mirror **1300**, a cluster **1400**, a center fascia **1500**, a passenger seat dashboard **1600**, and a display device **2**.

[0354] The side window glass **1100** and the front window glass **1200** may be partitioned by a pillar arranged between the side window glass **1100** and the front window glass **1200**.

[0355] The side window glass **1100** may be installed on a side of the vehicle **1000**. In an embodiment, the side window glass **1100** may be installed on a door of the vehicle **1000**. Multiple side window glasses **1100** may be provided and may face each other. In an embodiment, the side window glass **1100** may include a first side window glass **1110** and a second side window glass **1120**. In an embodiment, the first side window glass **1110** may be arranged adjacent to the cluster **1400**, and the second side window glass **1120** may be arranged adjacent to the passenger seat dashboard **1600**.

[0356] In an embodiment, the side window glasses **1100** may be spaced apart from each other in the x-direction or the −x-direction. For example, the first side window glass **1110** and the second side window glass **1120** may be spaced apart from each other in the x direction or the −x direction. For example, a virtual straight line L connecting the side window glasses **1100** may extend in the x-direction or the −x-direction. In an embodiment, a virtual straight line L connecting the first side window glass **1110** and the second side window glass **1120** to each other may extend in the x direction or the −x direction.

[0357] The front window glass **1200** may be installed in the front of the vehicle **1000**. The front window glass **1200** may be arranged between the side window glasses **1100** facing each other.

[0358] The side mirror **1300** may provide a rear view of the vehicle **1000**. The side mirror **1300** may be installed on the exterior of the vehicle body. In an embodiment, multiple side mirrors **1300** may be provided. For example, one of the side mirrors **1300** may be arranged outside the first side window glass **1110**, and another of the side mirrors **1300** may be arranged outside the second side window glass **1120**.

[0359] The cluster **1400** may be arranged in front of the steering wheel. The cluster **1400** may include a tachometer, a speedometer, a coolant thermometer, a fuel gauge, a turn signal indicator, a high beam indicator, a warning light, a seat belt warning light, an odometer, a tachograph, an automatic shift selector indicator light, a door open warning light, an engine oil warning light, and/or a low fuel warning light.

[0360] The center fascia **1500** may include a control panel on which buttons for adjusting an audio device, an air conditioning device, and a seat heater are arranged. The center fascia **1500** may be arranged on a side of the cluster **1400**.

[0361] The passenger seat dashboard **1600** may be spaced apart from the cluster **1400**, and the center fascia **1500** may be arranged between the cluster **1400** and the passenger seat dashboard **1600**. In an embodiment, the cluster **1400** may be arranged to correspond to a driver seat (not shown), and the passenger seat dashboard **1600** may be arranged to correspond to a passenger seat (not shown). In an embodiment, the cluster **1400** may be adjacent to the first side window glass **1110**, and the passenger seat dashboard **1600** may be adjacent to the second side window glass **1120**.

[0362] In an embodiment, the display device **2** may include a display panel **3**, and the display panel **3** may display an image. The display device **2** may be arranged inside the vehicle **1000**. In an embodiment, the display device **2** may be arranged between the side window glasses **1100** facing each other. The display device **2** may be arranged on at least one of the cluster **1400**, the center fascia **1500**, and the passenger seat dashboard **1600**.

[0363] The display device **2** may include an organic light-emitting display, an inorganic light-emitting display, a quantum dot display, or the like. Hereinafter, an organic light-emitting display device including the light-emitting device according to an embodiment will be described as an example of the display device **2**. However, various types of display devices as described above may be used in embodiments.

[0364] Referring to FIG. **7**A, the display device **2** may be arranged on the center fascia **1500**. In an embodiment, the display device **2** may display navigation information. In an embodiment, the display device **2** may display information regarding audio, video, or vehicle settings.

[0365] Referring to FIG. **7**B, the display device **2** may be arranged on the cluster **1400**. When the display device **2** is arranged on the cluster **1400**, the cluster **1400** may display driving information and the like through the display device **2**. For example, the cluster **1400** may digitally implement driving information and the like. The digital cluster **1400** may display vehicle information and driving information as images. For example, a needle and a gauge of a tachometer and various warning light or icons may be displayed by a digital signal.

[0366] Referring to FIG. **7**C, the display device **2** may be arranged on the passenger seat dashboard **1600**. The display device **2** may be embedded in the passenger seat dashboard **1600** or arranged on the passenger seat dashboard **1600**. In an embodiment, the display device **2** arranged on the passenger seat dashboard **1600** may display an image related to information displayed on the cluster **1400** and/or information displayed on the center fascia **1500**. In an embodiment, the display device **2** arranged on the passenger seat dashboard **1600** may display information that is different from information displayed on the cluster **1400** and/or information displayed on the center fascia **1500**.

[Manufacturing Method]

[0367] Respective layers included in the hole transport region **120**, the emission layer **130**, and respective layers included in the electron transport region **140** may be formed in a certain region by using various methods such as vacuum deposition, spin coating, casting, a Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, laser-induced thermal imaging (LITI), and the like.

[0368] When respective layers included in the hole transport region **120**, the emission layer **130**, and respective layers included in the electron transport region **140** are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10.sup.−8 torr to about 10.sup.−3 torr, and a deposition rate of about 0.01 Å/sec to about 100 Å/sec, depending on a material to be included in a layer to be formed and the structure of a layer to be formed.

Definitions of Terms

[0369] The term "C.sub.3-C.sub.60 carbocyclic group" as used herein may be a cyclic group consisting of carbon atoms as the only ring-forming atoms and having 3 to 60 carbon atoms.

[0370] The term "C.sub.1-C.sub.60 heterocyclic group" as used herein may be a cyclic group that has 1 to 60 carbon atoms and further includes, in addition to a carbon atom, at least one heteroatom as a ring-forming atom.

[0371] The C.sub.3-C.sub.60 carbocyclic group and the C.sub.1-C.sub.60 heterocyclic group may each be a monocyclic group consisting of one ring or a polycyclic group in which two or more rings are condensed with each other. For example, a C.sub.1-C.sub.60 heterocyclic group may have 3 to 61 ring-forming atoms.

[0372] The term "cyclic group" as used herein may be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group.

[0373] The term "π electron-rich C.sub.3-C.sub.60 cyclic group" as used herein may be a cyclic group that has 3 to 60 carbon atoms and may not include *—N═*′ as a ring-forming moiety.

[0374] The term "π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group" as used herein may be a heterocyclic group that has 1 to 60 carbon atoms and may include *—N═*′ as a ring-forming moiety.

[0375] In embodiments, [0376] a C.sub.3-C.sub.60 carbocyclic group may be a T1 group or a group in which two or more T1 groups are condensed with each other (for example, a cyclopentadiene group, an adamantane group, a norbornane group, a benzene group, a pentalene group, a naphthalene group, an azulene group, an indacene group, an acenaphthylene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a perylene group, a pentaphene group, a heptalene group, a naphthacene group, a picene group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, an indenophenanthrene group, or an indenoanthracene group), [0377] a C.sub.1-C.sub.60 heterocyclic group may be a T2 group, a group in which two or more T2 groups are condensed with each other, or a group in which one or more T2 groups and one or more T1 groups are condensed with each other (for example, a pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, a xanthene group, etc.), [0378] a π electron-rich C.sub.3-C.sub.60 cyclic group may be a T1 group, a group in which two or more T1 groups are condensed with each other, a T3 group, a group in which two or more T3 groups are condensed with each other, or a group in which one or more T3 groups and one or more T1 groups are condensed with each other (for example, a C.sub.3-C.sub.60 carbocyclic group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, etc.), and [0379] a π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group may be a T4 group, a group in which two or more T4 groups are condensed with each other, a group in which one or more T4 groups and one or more T1 groups are condensed with each other, a group in which one or more T4 groups and one or more T3 groups are condensed with each other, or a group in which one or more T4 groups, one or more T1 groups, and one or more T3 groups are condensed with one another (for example, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.), wherein [0380] a T1 group may be a cyclopropane group, a cyclobutane group, a cyclopentane group, a cyclohexane group, a cycloheptane group, a cyclooctane group, a cyclobutene group, a cyclopentene group, a cyclopentadiene group, a cyclohexene group, a cyclohexadiene group, a cycloheptene group, an adamantane group, a norbornane (or bicyclo[2.2.1]heptane) group, a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.2]octane group, or a benzene group, [0381] a T2 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a tetrazine group, a pyrrolidine group, an imidazolidine group, a dihydropyrrole group, a piperidine group, a tetrahydropyridine group, a dihydropyridine group, a hexahydropyrimidine group, a tetrahydropyrimidine group, a dihydropyrimidine group, a piperazine group, a tetrahydropyrazine group, a dihydropyrazine group, a tetrahydropyridazine group, or a dihydropyridazine group, [0382] a T3 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, or a borole group, and [0383] a T4 group may be a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or a tetrazine group.

[0384] The terms “cyclic group”, “C.sub.3-C.sub.60 carbocyclic group”, “C.sub.1-C.sub.60 heterocyclic group”, “π electron-rich C.sub.3-C.sub.60 cyclic group”, and “π electron-deficient

nitrogen-containing C.sub.1-C.sub.60 cyclic group" as used herein may each be a group condensed to any cyclic group, a monovalent group, or a polyvalent group (for example, a divalent group, a trivalent group, a tetravalent group, etc.) according to the structure of a formula for which the corresponding term is used.
[0385] For example, a "benzene group" may be a benzo group, a phenyl group, a phenylene group, or the like, which may be readily understood by one of ordinary skill in the art according to the structure of a formula including the "benzene group."
[0386] Examples of monovalent C.sub.3-C.sub.60 carbocyclic group or a monovalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group.
[0387] Examples of a divalent C.sub.3-C.sub.60 carbocyclic group or a divalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkylene group, a C.sub.1-C.sub.10 heterocycloalkylene group, a C.sub.3-C.sub.10 cycloalkenylene group, a C.sub.1-C.sub.10 heterocycloalkenylene group, a C.sub.6-C.sub.60 arylene group, a C.sub.1-C.sub.60 heteroarylene group, a divalent non-aromatic condensed polycyclic group, and a divalent non-aromatic condensed heteropolycyclic group.
[0388] The term "C.sub.1-C.sub.60 alkyl group" as used herein may be a linear or branched monovalent aliphatic hydrocarbon group that has 1 to 60 carbon atoms, and examples thereof may include a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, a tert-decyl group, and the like.
[0389] The term "C.sub.1-C.sub.60 alkylene group" as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.60 alkyl group.
[0390] The term "C.sub.2-C.sub.60 alkenyl group" as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon double bond in the middle or at a terminus of a C.sub.2-C.sub.60 alkyl group, and examples thereof may include an ethenyl group, a propenyl group, a butenyl group, and the like.
[0391] The term "C.sub.2-C.sub.60 alkenylene group" as used herein may be a divalent group having a same structure as the C.sub.2-C.sub.60 alkenyl group.
[0392] The term "C.sub.2-C.sub.60 alkynyl group" as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon triple bond in the middle or at a terminus of a C.sub.2-C.sub.60 alkyl group, and examples thereof may include an ethynyl group, a propynyl group, and the like.
[0393] The term "C.sub.2-C.sub.60 alkynylene group" as used herein may be a divalent group having a same structure as the C.sub.2-C.sub.60 alkynyl group.
[0394] The term "C.sub.1-C.sub.60 alkoxy group" as used herein may be a monovalent group represented by —O(A.sub.101) (wherein A.sub.101 may be a C.sub.1-C.sub.60 alkyl group), and examples thereof may include a methoxy group, an ethoxy group, an isopropyloxy group, and the like.
[0395] The term "C.sub.3-C.sub.10 cycloalkyl group" as used herein may be a monovalent saturated hydrocarbon cyclic group having 3 to 10 carbon atoms, and examples thereof may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a

cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group (or bicyclo[2.2.1]heptyl group), a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, and the like.

[0396] The term "C.sub.3-C.sub.10 cycloalkylene group" as used herein may be a divalent group having a same structure as the C.sub.3-C.sub.10 cycloalkyl group.

[0397] The term "C.sub.1-C.sub.10 heterocycloalkyl group" as used herein may be a monovalent cyclic group having 1 to 10 carbon atoms, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms, and examples thereof may include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, a tetrahydrothiophenyl group, and the like.

[0398] The term "C.sub.1-C.sub.10 heterocycloalkylene group" as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.10 heterocycloalkyl group.

[0399] The term "C.sub.3-C.sub.10 cycloalkenyl group" as used herein may be a monovalent cyclic group that has 3 to 10 carbon atoms, at least one carbon-carbon double bond in the cyclic structure thereof, and no aromaticity, and examples thereof may include a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, and the like.

[0400] The term "C.sub.3-C.sub.10 cycloalkenylene group" as used herein may be a divalent group having a same structure as the C.sub.3-C.sub.10 cycloalkenyl group.

[0401] The term "C.sub.1-C.sub.10 heterocycloalkenyl group" as used herein may be a monovalent cyclic group that has 1 to 10 carbon atoms, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms, and at least one carbon-carbon double bond in the cyclic structure thereof. Examples of a C.sub.1-C.sub.10 heterocycloalkenyl group may include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, a 2,3-dihydrothiophenyl group, and the like.

[0402] The term "C.sub.1-C.sub.10 heterocycloalkenylene group" as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.10 heterocycloalkenyl group.

[0403] The term "C.sub.6-C.sub.60 aryl group" as used herein may be a monovalent group having a carbocyclic aromatic system of 6 to 60 carbon atoms.

[0404] The term "C.sub.6-C.sub.60 arylene group" as used herein may be a divalent group having a carbocyclic aromatic system of 6 to 60 carbon atoms.

[0405] Examples of a C.sub.6-C.sub.60 aryl group may include a phenyl group, a pentalenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a heptalenyl group, a naphthacenyl group, a picenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, and the like.

[0406] When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the respective two or more rings may be condensed with each other.

[0407] The term "C.sub.1-C.sub.60 heteroaryl group" as used herein may be a monovalent group having a heterocyclic aromatic system of 1 to 60 carbon atoms, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms.

[0408] The term "C.sub.1-C.sub.60 heteroarylene group" as used herein may be a divalent group having a heterocyclic aromatic system of 1 to 60 carbon atoms, and further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms.

[0409] Examples of a C.sub.1-C.sub.60 heteroaryl group may include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, a benzoquinolinyl group, an isoquinolinyl group, a benzoisoquinolinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthrolinyl group, a phthalazinyl group, a naphthyridinyl group, and the like.

[0410] When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the respective two or more rings may be condensed with each

other.
[0411] The term "monovalent non-aromatic condensed polycyclic group" as used herein may be a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed to each other, only carbon atoms as ring-forming atoms, and no aromaticity in its molecular structure as a whole. Examples of a monovalent non-aromatic condensed polycyclic group may include an indenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, an indenophenanthrenyl group, an indeno anthracenyl group, and the like.
[0412] The term "divalent non-aromatic condensed polycyclic group" as used herein may be a divalent group having a same structure as the monovalent non-aromatic condensed polycyclic group.
[0413] The term "monovalent non-aromatic condensed heteropolycyclic group" as used herein may be a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms, and no aromaticity in its molecular structure as a whole. Examples of a monovalent non-aromatic condensed heteropolycyclic group may include a pyrrolyl group, a thiophenyl group, a furanyl group, an indolyl group, a benzoindolyl group, a naphthoindolyl group, an isoindolyl group, a benzoisoindolyl group, a naphthoisoindolyl group, a benzosilolyl group, a benzothiophenyl group, a benzofuranyl group, a carbazolyl group, a dibenzosilolyl group, a dibenzothiophenyl group, a dibenzofuranyl group, an azacarbazolyl group, an azafluorenyl group, an azadibenzosilolyl group, an azadibenzothiophenyl group, an azadibenzofuranyl group, a pyrazolyl group, an imidazolyl group, a triazolyl group, a tetrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, an oxadiazolyl group, a thiadiazolyl group, a benzopyrazolyl group, a benzimidazolyl group, a benzoxazolyl group, a benzothiazolyl group, a benzoxadiazolyl group, a benzothiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an imidazotriazinyl group, an imidazopyrazinyl group, an imidazopyridazinyl group, an indenocarbazolyl group, an indolocarbazolyl group, a benzofurocarbazolyl group, a benzothienocarbazolyl group, a benzosilolocarbazolyl group, a benzoindolocarbazolyl group, a benzocarbazolyl group, a benzonaphthofuranyl group, a benzonaphthothiophenyl group, a benzonaphthosilolyl group, a benzofurodibenzofuranyl group, a benzofurodibenzothiophenyl group, a benzothienodibenzothiophenyl group, and the like.
[0414] The term "divalent non-aromatic condensed heteropolycyclic group" as used herein may be a divalent group having a same structure as the monovalent non-aromatic condensed heteropolycyclic group.
[0415] The term "C.sub.6-C.sub.60 aryloxy group" as used herein may be a group represented by —O(A.sub.102) (wherein A.sub.102 may be a C.sub.6-C.sub.60 aryl group).
[0416] The term "C.sub.6-C.sub.60 arylthio group" as used herein may be a group represented by —S(A.sub.103) (wherein A.sub.103 may be a C.sub.6-C.sub.60 aryl group).
[0417] The term "C.sub.7-C.sub.60 arylalkyl group" as used herein may be a group represented by -(A.sub.104)(A.sub.105) (wherein A.sub.104 may be a C.sub.1-C.sub.54 alkylene group, and A.sub.105 may be a C.sub.6-C.sub.59 aryl group).
[0418] The term "C.sub.2-C.sub.60 heteroarylalkyl group" as used herein may be a group represented by -(A.sub.106)(A.sub.107) (wherein A.sub.106 may be a C.sub.1-C.sub.59 alkylene group, and A.sub.107 may be a C.sub.1-C.sub.59 heteroaryl group).
[0419] In the specification, the group R.sub.10a may be: [0420] deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; [0421] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —

N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(═O)(Q.sub.11), —S(═O).sub.2(Q.sub.11), —P(═O)(Q.sub.11)(Q.sub.12), or any combination thereof; [0422] a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, or a C.sub.2-C.sub.60 heteroarylalkyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(═O)(Q.sub.21), —S(═O).sub.2(Q.sub.21), —P(═O)(Q.sub.21)(Q.sub.22), or any combination thereof; or [0423] —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), or —P(═O)(Q.sub.31)(Q.sub.32).
[0424] In the specification, Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; C.sub.1-C.sub.60 alkyl group; C.sub.2-C.sub.60 alkenyl group; C.sub.2-C.sub.60 alkynyl group; C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.
[0425] The term “heteroatom” as used herein may be any atom other than a carbon atom or a hydrogen atom. Examples of a heteroatom may include O, S, N, P, Si, B, Ge, Se, or any combinations thereof.
[0426] In the specification, examples of a “third-row transition metal” may include hafnium (Hf), tantalum (Ta), tungsten (W), rhenium (Re), osmium (Os), iridium (Ir), platinum (Pt), gold (Au), and the like.
[0427] In the specification, the term “D” may refer to deuterium, the term “Ph” may refer to a phenyl group, the term “Me” may refer to a methyl group, the term “Et” may refer to an ethyl group, the terms “tert-Bu”, “.sup.tBu,” and “Bu.sup.t” may each refer to a tert-butyl group, and the term “OMe” may refer to a methoxy group.
[0428] The term “biphenyl group” as used herein may be a “phenyl group substituted with a phenyl group.” For example, a “biphenyl group” may be a substituted phenyl group having a C.sub.6-C.sub.60 aryl group as a substituent.
[0429] The term “terphenyl group” as used herein may be a “phenyl group substituted with a biphenyl group”. For example, a “terphenyl group” may be a “substituted phenyl group” wherein the substituent is a “C.sub.6-C.sub.60 aryl group substituted with a C.sub.6-C.sub.60 aryl group”, or a “terphenyl group” may be a “substituted phenyl group” wherein two substituents are present, and each substituent is a “C.sub.6-C.sub.60 aryl group.”
[0430] In the specification, the symbols * and *′, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula or moiety.
[0431] In the specification, the terms “x-axis”, “y-axis”, and “z-axis” are not limited to three axes in an orthogonal coordinate system (for example, a Cartesian coordinate system), and may be interpreted in a broader sense than the aforementioned axes in an orthogonal coordinate system. For example, the x-axis, y-axis, and z-axis may describe axes that are orthogonal to each other, or may describe axes that are in different directions that are not orthogonal to each other.
[0432] Hereinafter, an organic compound according to an embodiment and a light-emitting device according to an embodiment will be described in further detail with reference to the Examples and the Comparative Examples.
EXAMPLES
Evaluation Example 1 (Evaluation of Photoluminescence Quantum Yields (PLQY) and Triplet

Exciton Lifetime)

[0433] A thin-film with a thickness of 400 Å was formed by using Compound HTH2 as a hole transport host, Compound ETH2 as an electron transport host, and the organic compounds listed in Table 1 below at a weight ratio of 49:49:2. For each thin-film, PLQY, delayed fluorescence photoluminescence quantum yield (ϕ.sub.DF), prompt fluorescence photoluminescence quantum yield (ϕ.sub.PF), and triplet exciton lifetime (Td) were measured from the photoluminescence (PL) spectrum measured at a temperature of 300 K by using PL measurement equipment, and the results are shown in Table 1 below. PLQY was calculated by measuring the photoluminescence quantum yields at an excitation wavelength in a range of 280 nm to 320 nm using an integrating sphere and by taking the average value thereof. The delayed fluorescence photoluminescence quantum yield and the prompt fluorescence photoluminescence quantum yield relative to a total PLQY were calculated from the amplitudes of the first and second components of the transient PL (TRPL) decay curve. In this regard, the total PLQY is equal to a sum of the delayed fluorescence photoluminescence quantum yield and the prompt fluorescence photoluminescence quantum yield.

TABLE-US-00001 TABLE 1

Organic compound	PLQY (%)	Φ.sub.DF (%)	Φ.sub.PF (%)	Triplet exciton lifetime (τ.sub.d) (μs)
DF1	95	45	50	80
DF2	96	45	51	88
DF3	96	47	49	60
DF4	90	44	46	88
DF5	88	44	44	85
DFD13	85	7	78	5
DFD14	96	9	87	8

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[0434] From Table 1, it can be seen that in each of Compounds DFD13 and DFD14, which are organic compounds appropriately having both the short charge transfer (short CT) characteristics and the long charge transfer (long CT) characteristics by appropriately including both a portion where the distance between the HOMO region and the LUMO region is relatively close and a portion where the distance between the HOMO region and the LUMO region is relatively long, the ϕ.sub.PF/(ϕ.sub.PF+ϕ.sub.DF) value is larger and the triplet exciton lifetime is smaller than in Compounds DF1 to DF5.

Evaluation Example 2 (Evaluation of ΔE.SUB.ST .and Oscillator Strength f Value)

[0435] The PL spectrum of each sample in which each of the organic compounds listed in Table 2 below was dissolved in toluene at a concentration of 10.sup.−5 M was measured at a temperature of 300 K, and the PL spectrum of each sample in which the respective organic compound was dissolved in tetrahydrofuran at a concentration of 10.sup.−5 M was measured at a temperature of 77 K. For each of the organic compounds listed in Table 2 below, ΔE.sub.ST was calculated as the difference in the onsets of the two PL spectra and the results are shown in Table 2 below.

[0436] The oscillator strength f value was calculated by using Gaussian16 simulation and the results are shown in Table 2 below.

TABLE-US-00002 TABLE 2

Organic compound	DF1	DF2	DF3	DF4	DF5	DFD13	DFD14
ΔE.sub.ST (eV)	0.15	0.2	0.18	0.18	0.16	0.13	0.12
Oscillator strength (f)	0.4800	0.4656	0.3655	0.4101	0.4299	0.3250	0.3890

[0437] From Table 2, it can be seen that each of Compounds DFD13 and DFD14, which are organic compounds appropriately having both short CT and long CT characteristics, has a similar level of oscillator strength as in Compounds DF1 to DF5, but is suitable for being used as a thermally activated delayed fluorescence (TADF) material as it has a lower ΔE.sub.ST compared to Compounds DF1 to DF5.

Evaluation Example 3 (Evaluation of k.SUB.r..SUP.S., k.SUB.ISC., and k.SUB.RISC.)

[0438] For each organic compound listed in Table 3 below, k.sub.r.sup.S was calculated by using the reciprocal of the singlet exciton lifetime (τ.sub.p), k.sub.ISC was calculated by using Equation

[00011] $\frac{1 - \varnothing_{PF}}{p}$,

and k.sub.RISC was calculated by using Equation

[00012] $\frac{\varnothing_{DF}}{k_{ISC}\ \ p\ \ d\ \varnothing_{PF}}$

(wherein ϕ.sub.PF is a prompt fluorescence photoluminescence quantum yield (PF PLQY), and ϕ.sub.DF is a delayed fluorescence photoluminescence quantum yield (DF PLQY)). The results thereof are shown in Table 3 below.

TABLE-US-00003 TABLE 3 Organic compound k.sub.r.sup.S (10.sup.7s.sup.−1) k.sub.ISC (10.sup.6s.sup.−1) K.sub.RISC (10.sup.4s.sup.−1) DF1 11 11 6 DF2 5 8 5 DF3 18 11 6 DF4 18 12 6 DF5 14 8 4 DFD13 8 1 24 DFD14 9 3 38

[0439] From Table 3, it can be seen that each of Compounds DFD13 and DFD14, which are organic compounds appropriately having both short CT and long CT characteristics, has a relatively large k.sub.RISC compared to Compounds DF1 to DF5, thereby satisfying Expression 3 as described above, and has a relatively small k.sub.ISC compared to Compounds DF1 to DF5, thereby satisfying Expression 1 as described above.

Comparative Example 1

[0440] A glass substrate (made by Corning) with an ITO electrode of 15 Ω/cm.sup.2 (1200 Å) as an anode was cut into pieces, each of which has a size of 50 mm×50 mm×0.7 mm, and each piece was ultrasonically cleaned for 5 minutes by using isopropyl alcohol and pure water, cleaned by exposure to ozone and with irradiation of ultraviolet rays thereto for 30 minutes, and mounted in a vacuum deposition device.

[0441] NPD was vacuum deposited on the anode to form a hole injection layer with a thickness of 300 Å. Compound HT3 was vacuum deposited on the hole injection layer to form a hole transport layer with a thickness of 200 Å. CzSi was vacuum deposited on the hole transport layer to form an electron blocking layer with a thickness of 100 Å.

[0442] An emission layer with a thickness of 200 Å was formed by vacuum co-depositing a host mixture of Compound ETH2 (electron transport host) and Compound HTH17 (hole transport host) mixed at a ratio of 1:1, Compound PD39 (sensitizer), and Compound DF1 (organic compound, dopant) on the electron blocking layer at a host:sensitizer:dopant weight ratio of 85:14:1.

[0443] TSPO1 was vacuum deposited on the emission layer to form a hole blocking layer with a thickness of 200 Å. TPBi was vacuum deposited on the hole blocking layer to form an electron transport layer with a thickness of 300 Å. An electron injection layer with a thickness of 10 Å was formed by vacuum depositing LiF on the electron transport layer, and Al was vacuum deposited thereon to form a cathode with a thickness of 3000 Å, thus forming a LiF/Al electrode and thereby completing the manufacture of a light-emitting device.

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Comparative Examples 2 to 5 and Examples 1 and 2

[0444] A light-emitting device was manufactured by using the same method as Comparative Example 1, except that the organic compounds listed in Table 4 below were used instead of Compound DF1 when forming the emission layer.

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Evaluation Example 4 (Evaluation of Luminescence Efficiency and Lifespan of Light-Emitting Device)

[0445] The luminescence efficiency (Cd/A) and lifespan (LT.sub.95, hr) of the light-emitting device manufactured respectively in Comparative Examples 1 to 5 and Examples 1 and 2 were measured at a current density of 10 mA/cm.sup.2 by using the V7000 OLED IVL Test System (Polaronix). The lifespan (LT.sub.95) refers to the time (hr) it takes for the luminance to reach 95% of the initial luminance. The measured results are shown in Table 4 below as relative values compared to Comparative Example 1. Whether Expressions 1 to 8 are satisfied or not based on the results according to the Evaluation Examples 1 to 3 (satisfied: O, dissatisfied: X) is shown in Table 4 below, and the value within parentheses refers to a value for each corresponding Expression.

TABLE-US-00004 TABLE 4 Comparative Comparative Example 3 Example 4 Example 5 Example 1 Example 2 Comparative Comparative Comparative Example 1 Example 2 Organic DF1 DF2 DF3 DF4 DF5 DFD13 DFD14 compound Expression X X X X X O O 1 (183.3) (160) (183.3) (200) (200) (4.2) (7.9) Expression O O O O O O O 2 (11 × 10.sup.6 s.sup.−1) (8 × 10.sup.6 s.sup.

−1) (11 × 10.sup.6 s.sup.−1) (12 × 10.sup.6 s.sup.−1) (8 × 10.sup.6 s.sup.−1) (10.sup.6 s.sup.−1) (3 × 10.sup.6 s.sup.−1) Expression X X X X X O O 3 (6 × 10.sup.4 s.sup.−1) (5 × 10.sup.4 s.sup.−1) (6 × 10.sup.4 s.sup.−1) (6 × 10.sup.4 s.sup.−1) (4 × 10.sup.4 s.sup.−1) (24 × 10.sup.4 s.sup.−1) (38 × 10.sup.4 s.sup.−1) Expression O O O O O O O 4 (11 × 10.sup.7 s.sup.−1 >| (5 × 10.sup.7 s.sup. −1 > (18 × 10.sup.7 s.sup.−1 > (18 × 10.sup.7 s.sup.−1 > (14 × 10.sup.7 s.sup.−1 > (8 × 10.sup.7 s.sup.−1 > (9 × 10.sup.7 s.sup.−1 > 11 × 10.sup.6 s.sup.−1) 8 × 10.sup.6 s.sup.−1) 11 × 10.sup.6 s.sup.−1) 12 × 10.sup.6 s.sup.−1) 8 × 10.sup.6 s.sup.−1) 1 × 10.sup.6 s.sup.−1) 3 × 10.sup.6 s.sup. −1) Expression O O O O O O O 5 (11 × 10.sup.7 s.sup.−1 > (5 × 10.sup.7 s.sup.−1 > (18 × 10.sup.7 s.sup.−1 > (18 × 10.sup.7 s.sup.−1 > (14 × 10.sup.7 s.sup.−1 > (8 × 10.sup.7 s.sup.−1 > (9 × 10.sup.7 s.sup.−1 > 6 × 10.sup.4 s.sup.−1) 5 × 10.sup.4 s.sup.−1) 6 × 10.sup.4 s.sup.−1) 6 × 10.sup.4 s.sup.−1) 4 × 10.sup.4 s.sup.−1) 24 × 10.sup.4 s.sup.−1) 38 × 10.sup.4 s.sup.−1) Expression O O O O O O O 6 (11 × 10.sup.7 s.sup.−1) (5 × 10.sup.7 s.sup.−1) (18 × 10.sup.7 s.sup.−1) (18 × 10.sup.7 s.sup.−1) (14 × 10.sup.7 s.sup.−1) (8 × 10.sup.7 s.sup.−1) (9 × 10.sup.7 s.sup.−1) Expression X X X X X O O 7 (52.6%) (53.1%) (51.0%) (51.1%) (50.0%) (91.8%) (90.6%) Expression O X X X X O O 8 (0.15 eV) (0.2 eV) (0.18 eV) (0.18 eV) (0.16 eV) (0.13 eV) (0.12 eV) Luminescence 100 96 96 90 88 85 96 efficiency (%) Lifespan 100 88 75 78 65 127 129 (%)

[0446] From Table 4, it can be confirmed that Compounds DFD13 and DFD14, which are organic compounds appropriately having both short CT and long CT characteristics, satisfy Expressions 1 to 8, and the light-emitting devices of Examples 1 and 2 respectively using Compounds DFD13 and DFD14 have luminescence efficiencies similar to those of the light-emitting devices of Comparative Examples 1 to 5 respectively using Compounds DF1 to DF5 that did not satisfy at least Expression 1, while also having excellent lifespans.

[0447] A boron-containing organic compound that appropriately has both short CT and long CT characteristics and thus satisfies at least Expression 1 of Expressions 1 to 8 as described above may have a relatively high k.sub.r.sup.S and a relatively high k.sub.RISC compared to a boron-containing compound not satisfying Expression 1, while also having an oscillator strength f value similar to that of the boron-containing compound not satisfying Expression 1. Therefore, when a boron-containing organic compound satisfies Expression 1, triplet excitons may be harvested into singlet excitons relatively quickly, and the singlet excitons may be relatively quickly moved to the ground state and emit light.

[0448] Embodiments have been disclosed herein, and although terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for the purposes of limitation. In some instances, as would be apparent by one of ordinary skill in the art, features, characteristics, and/or elements described in connection with an embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of ordinary skill in the art that various changes in form and details may be made without departing from the spirit and scope of the disclosure as set forth in the claims.

Claims

1. A light-emitting device comprising: a first electrode; a second electrode facing the first electrode; and an interlayer between the first electrode and the second electrode and including an emission layer, wherein the interlayer comprises an organic compound that includes a boron (B) atom, and the organic compound satisfies Expression 1:
k.sub.ISC/*k*.sub.RISC≤10 [Expression 1] wherein in Expression 1, k.sub.ISC is an intersystem crossing rate constant of the organic compound, and k.sub.RISC is a reverse intersystem crossing rate constant of the organic compound.

2. The light-emitting device of claim 1, wherein the interlayer further comprises a hole transport host, an electron transport host, a sensitizer, or a combination thereof, and the organic compound, the hole transport host, the electron transport host, and the sensitizer are each different from each other.

3. The light-emitting device of claim 2, wherein the hole transport host and the electron transport host form an exciplex.

4. The light-emitting device of claim 2, wherein the hole transport host is a compound comprising at least one carbazole group.

5. The light-emitting device of claim 2, wherein the electron transport host is a compound comprising at least one π electron-deficient nitrogen-containing 6-membered ring.

6. The light-emitting device of claim 2, wherein the sensitizer comprises a transition metal.

7. The light-emitting device of claim 2, wherein the emission layer comprises: the organic compound; and the hole transport host, the electron transport host, the sensitizer, or a combination thereof.

8. The light-emitting device of claim 1, wherein the emission layer emits blue light having a maximum emission wavelength in a range of about 450 nm to about 470 nm.

9. An electronic apparatus comprising the light-emitting device of claim 1.

10. The electronic apparatus of claim 9, further comprising: a thin-film transistor electrically connected to the light-emitting device.

11. The electronic apparatus of claim 9, further comprising: a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or a combination thereof.

12. An electronic equipment comprising the light-emitting device of claim 1, wherein the electronic equipment is a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.

13. An organic compound including a boron (B) atom, wherein the organic compound satisfies Expression 1: $k_{\mathrm{ISC}} / k_{\mathrm{RISC}} \leq 10$ [Expression1] wherein in Expression 1, k.sub.ISC is an intersystem crossing rate constant of the organic compound, and k.sub.RISC is a reverse intersystem crossing rate constant of the organic compound.

14. The organic compound of claim 13, wherein the organic compound further satisfies at least one of Expressions 2 and 3: $k_{\mathrm{ISC}} \geq 10^{5} s^{-1}$ [Expression2] $k_{\mathrm{RISC}} \geq 10^{5} s^{-1}$. [Expression3]

15. The organic compound of claim 13, wherein the organic compound further satisfies at least one of Expressions 4 and 5: $k_r^S > k_{\mathrm{ISC}}$ [Expression4] $k_r^S > k_{\mathrm{RISC}}$ [Expression5] wherein in Expressions 4 and 5, k.sub.r.sup.S is a radiative rate constant of a lowest singlet excited state of the organic compound.

16. The organic compound of claim 13, wherein the organic compound further satisfies Expression 6: $k_r^S \geq 10^{7} s^{-1}$ [Expression6] wherein in Expression 6, k.sub.r.sup.S is a radiative rate constant of a lowest singlet excited state of the organic compound.

17. The organic compound of claim 13, wherein the organic compound further satisfies Expression 7: $_{\mathrm{PF}} / (_{\mathrm{PF}} + _{\mathrm{DF}}) \geq 90\%$ [Expression7] wherein in Expression 7, ϕ.sub.PF is a prompt fluorescence photoluminescence quantum yield (PF PLQY) of the organic compound, and ϕ.sub.DF is a delayed fluorescence photoluminescence quantum yield (DF PLQY) of the organic compound.

18. The organic compound of claim 13, wherein the organic compound further satisfies Expression

8: $E_{ST} \leq 0.15eV$ [Expression8] wherein in Expression 8, ΔE.sub.ST is a difference between a lowest singlet excited state (S.sub.1) energy level of the organic compound and a lowest triplet excited state (T.sub.1) energy level of the organic compound.

19. The organic compound of claim 13, wherein the organic compound further includes at least one nitrogen atom.

20. The organic compound of claim 13, wherein the organic compound comprises a heterocyclic group containing a boron atom and a nitrogen atom each as a ring-forming atom.